

The AASC–Combinatorics Endpoint Bridge

An Integrated Hybrid Closure for the Three-Petal Sunflower Theorem

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Abstract

Let $\mathcal{F}$ be a finite n -uniform family with no three-petal sunflower. The proof combines two complementary layers. Positive finite combinatorics constructs a cardinality-minimal blocker with private edges, the residual tower, every compatible continuation path, raw terminal records, finite type coordinates, collision dispositions, and a well-founded certified terminalization tree. Genuine many-to-one residualization is retained: the rank-two triangle is a permanent regression against unrestricted pre-discharge root-to-seed injectivity.

At every generated collision, the proof assigns the difference to endpoint-neutral bookkeeping collapse, bounded same-side charge, tensor or scope separation, literal sunflower realization, or strict lower-rank reconstruction. After these branches have been discharged, the surviving fixed-scope package is certificate-terminal. The manuscript-specific AASC exhaustion/impossibility theorem then states that two independent standing-bearing endpoint occupants cannot remain at one fixed terminal locus. Kernel governance is not an additional sunflower-family hypothesis: determinate, non-degenerate terminal incidences instantiate Admissibility, Standing, Reference, and Irreversibility by necessity.

The terminal AASC relation and the independently defined combinatorial endpoint relation are retained. Their biconditional coincidence

$$a \equiv_{\text{term}} b \iff a \equiv_{\text{comb}} b$$

induces an endpoint-faithful equivalence

$$\text{AASCEnd}_{G,U} \simeq \text{CombEnd}_{G,U}.$$

This bridge identifies two faithful presentations of the same constraint-saturated endpoint; it does not itself manufacture injectivity. Terminal closeout supplies quotient-final seed rigidity. Original-edge coverage is part of the endpoint tensor, and private-edge minimality gives the stronger antichain law

$$\text{Cov}_{\mathcal{F}}(x) \subseteq \text{Cov}_{\mathcal{F}}(y) \iff x = y \quad (x, y \in B^{(0)}).$$

Consequently, coverage-preserving quotienting may remove duplicate histories over one root, but it cannot compress distinct coverage-bearing roots of the minimal blocker. Choosing one surviving terminal record for each source gives, after closeout, an injective source-to-seed map. Hence every exact source fibre satisfies $|U| \leq 4094$, and the actual minimal blocker satisfies

$$|B^{(0)}| \leq 8 \cdot 2^6 \cdot 4 \cdot 4094 = 8,384,512.$$

The ordinary one-point-link recurrence therefore yields

$$|\mathcal{F}| \leq (8,384,512)^n.$$

The accompanying Lean development formalizes the combinatorial and proof-theoretic component at its natural boundary: positive generation, real merge disposition, the triangle regression, coverage rigidity, certified terminalization, necessity-rooted kernel governance, elimination

of the attempted independent-authorizer branch, the four-role identity closeout, fixed identity, the support-fibre bound, and endpoint transfer. The manuscript proves the Sunflower-specific generated five-way collision classification and its literal reduction certificates at their natural boundary. A typed theorem import joins these completed components, after which Lean derives the terminal closeout and endpoint consequences. This is the complete integrated proof architecture. Replacing its terminal AASC theorem by a wholly autonomous conventional reconstruction would be a distinct strengthening, neither assumed nor claimed here.

Scope and dependency conventions

One hybrid proof. The manuscript and the Lean development are complementary parts of one argument. The manuscript supplies the Sunflower-specific generated five-way collision classification and literal reduction certificates. Lean supplies the generated combinatorics, necessity-rooted governance, fifth-branch elimination, four-role terminal closeout, fixed identity, coverage rigidity, and endpoint-transfer audit.

Positive generation is combinatorial. Track I generates blockers, private edges, residuals, local slots, every compatible path, raw terminal records, coverage tensors, finite type coordinates, and literal certificates. AASC does not generate a path, seed, selector, matching, or blocker.

Raw merges are lawful. Iterated residualization may be many-to-one. The triangle refutes unrestricted pre-discharge root-to-seed injectivity. It does not refute injectivity proved after every lawful collision branch has been discharged.

Every collision receives a disposition. A consequence-bearing difference enters bounded charge, tensor or scope separation, sunflower realization, or strict lower-rank reconstruction. Endpoint-neutral history or bookkeeping multiplicity collapses without changing the surviving endpoint.

Terminal closeout supplies occupancy rigidity. In a certificate-terminal, fixed-scope, non-degenerate determinate package, all lawful alternatives have been exhausted. Kernel necessity and fixed-domain AASC impossibility then exclude two independent standing-bearing occupants at one fixed terminal locus. Lean derives this closeout after importing only the target-specific generated witnesses and certificates.

The endpoint bridge is retained. The AASC terminal relation and the combinatorial endpoint relation are defined independently. Their biconditional coincidence identifies two faithful presentations of the same saturated endpoint and transports closeout between them.

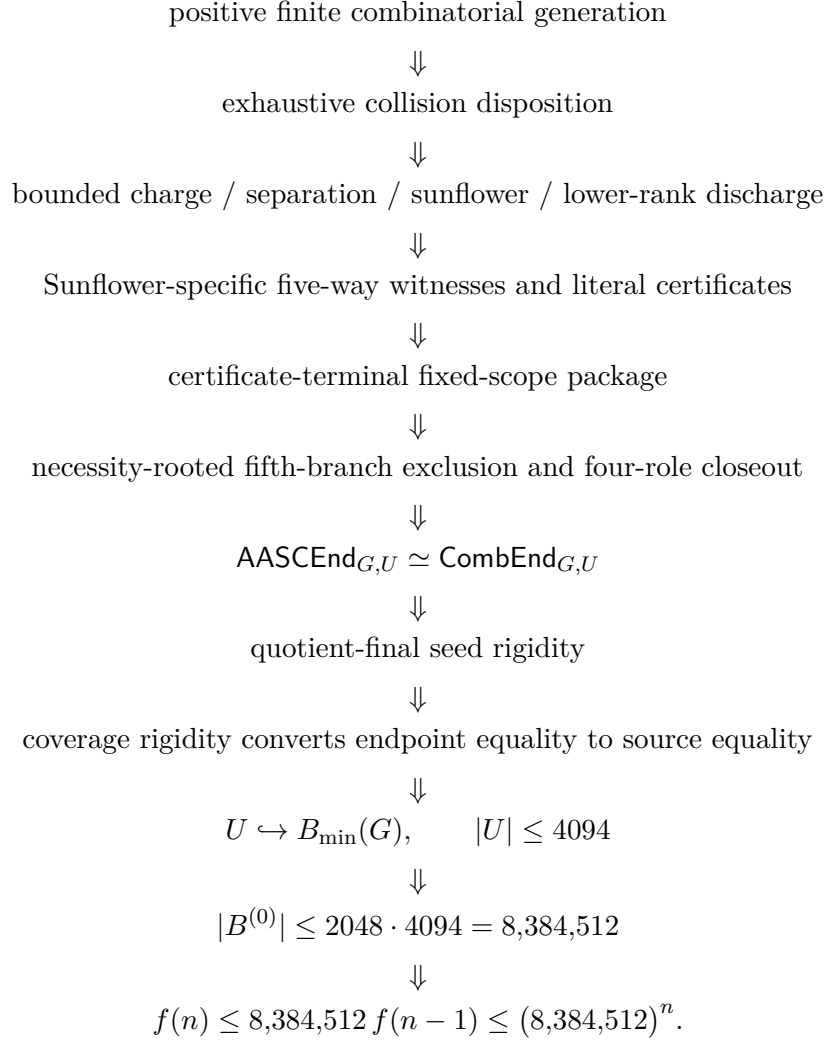
Quotients remove histories, not blocker roots. Endpoint-neutral route multiplicity and duplicate histories may collapse. Distinct roots of a cardinality-minimal blocker cannot be merged by any quotient that preserves original-edge coverage.

The actual source fibre is bounded. No inequality $|U| \leq 4094$ is assumed at the raw stage. After terminal closeout, quotient-final seed rigidity and coverage rigidity produce an injective map from the surviving source fibre into the 4094 terminal seeds.

Simultaneity is logical. Standing, Reference, tensor content, coverage, and licensed metric description are jointly instantiated at the saturated endpoint. The proof order records authorizing dependence, not temporal succession. Cardinality reports the endpoint population; it does not create primitive identity.

The recurrence uses the original minimal blocker. The bounded object is the actual cardinality-minimal blocker $B^{(0)}$, not a smaller quotient representative. The ordinary one-point-link recurrence is applied directly to $B^{(0)}$.

Controlling proof architecture



Audit Gate 0.1 (Prohibited reversals). *The proof never infers raw Hall from determinacy, never infers terminal closeout merely from endpoint equivalence, and never infers a smaller blocker from coverage-preserving quotienting. Seed rigidity appears only after certificate-terminal AASC closeout; source injectivity appears only after seed rigidity and minimal-blocker coverage rigidity.*

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1. Claim, endpoint objects, and dependency order

1.1. Uniform families and exact source fibres

Definition 1.1 (Uniform family and three-sunflower). *Let X be finite. A family $\mathcal{F} \subseteq \binom{X}{n}$ is n -uniform. Three distinct members $A_0, A_1, A_2 \in \mathcal{F}$ form a three-sunflower if there is a core C such that $A_i \cap A_j = C$ whenever $i \neq j$.*

Definition 1.2 (Fixed typed source fibre). *For a canonical support cell S , six-bit generated profile p , and generated coarse role r , put*

$$U_{S,p,r} = \{x \in B^{(0)} : \text{supp}(x) = S, \text{prof}(x) = p, \text{role}(x) = r\}.$$

The support, profile, and role coordinates are generated before terminal readout. They partition the actual cardinality-minimal blocker and contribute at most $8 \cdot 2^6 \cdot 4 = 2048$ exact source types.

Definition 1.3 (Raw terminal record population). *Let $G = \mathcal{F}^{(m)}$ be the controlled terminal residual family. For an exact source fibre U , define*

$$\text{RawTerm}_{G,U} = \{(x, \pi, z) : x \in U, \pi \in \text{Path}_m(x), \text{last}(\pi) = z \in B_{\min}(G)\}.$$

This finite population is positively generated by Track I. Several histories, and even histories from distinct roots, may reach one terminal coordinate. No uniqueness or Hall condition is built into the definition.

Definition 1.4 (Terminal probe family). *The admitted terminal probe family $\mathcal{H}_{G,U}^{\text{term}}$ consists of the already instantiated endpoint predicates on raw terminal records:*

1. *Admissibility, Standing, Reference, and Irreversibility status;*
2. *the terminal seed and its private-edge incidence signature;*
3. *endpoint-relevant continuation, truncation, reconstruction, and factorization status;*
4. *literal sunflower, closeout, split, target-change, and scope-change status;*
5. *the exact typed source coordinates (S, p, r) ; and*
6. *the original-edge coverage tensor defined in Section 5.*

Historical spelling, path-list order, provenance formatting, duplicate ledger entries, and raw presentation multiplicity are not independent probes when their variation changes none of these admitted consequences.

Definition 1.5 (AASC terminal standing equivalence). *For $a, b \in \text{RawTerm}_{G,U}$, write*

$$a \equiv_{\text{term}} b$$

when every admitted terminal Standing, Reference, tensor, coverage, and status probe has the same value on a and b . Define

$$\text{AASCEnd}_{G,U} = \text{TCar}_{G,U} = \text{RawTerm}_{G,U} / \equiv_{\text{term}}.$$

This is the AASC endpoint presentation. It is not quotienting by terminal seed equality alone.

Definition 1.6 (Combinatorial endpoint signature). *For $a = (x, \pi, z) \in \text{RawTerm}_{G,U}$, let $\text{csig}_{G,U}(a)$ be the generated recurrence-sufficient terminal signature consisting of:*

1. *the exact source type (S, p, r) ;*
2. *the terminal coordinate z and its private-edge incidence signature;*
3. *the original-edge coverage tensor $\text{TCov}_{G,U}(a)$;*
4. *endpoint-relevant continuation, truncation, reconstruction, and factorization data; and*
5. *the literal sunflower, closeout, split, target-change, and scope-change profile.*

It contains every generated combinatorial coordinate whose variation changes the surviving endpoint or recurrence. Endpoint-neutral history formatting is omitted.

Definition 1.7 (Combinatorial endpoint equivalence). *For $a, b \in \text{RawTerm}_{G,U}$, write*

$$a \equiv_{\text{comb}} b \iff \text{csig}_{G,U}(a) = \text{csig}_{G,U}(b),$$

and define

$$\text{CombEnd}_{G,U} = \text{RawTerm}_{G,U} / \equiv_{\text{comb}}.$$

The relations $\equiv_{\text{term}}$ and $\equiv_{\text{comb}}$ are independently specified. Their coincidence is a theorem proved only after terminal disposition and AASC closeout.

Definition 1.8 (Terminal seed readout). *For an endpoint class $C = [(x, \pi, z)]$, write*

$$\bar{\epsilon}_{G,U}(C) = z$$

when this value is read in either endpoint presentation. Section 5 proves well-definedness and quotient-final injectivity on the certificate-terminal package. The readout records the already fixed terminal locus; it does not create endpoint identity.

Theorem 1.9 (Three-petal constant-base endpoint). *Every finite n -uniform three-sunflower-free family satisfies*

$$|\mathcal{F}| \leq (8,384,512)^n.$$

Equivalently, every finite n -uniform family of cardinality greater than $(8,384,512)^n$ contains a three-sunflower.

1.2. Exact division of labour

Track I proves the minimal-blocker and private-edge construction, residual-fibre control, residual-plus-slot injectivity, branch-complete path population, finite constructor saturation, raw terminal records, generated type coordinates, the exhaustive collision-disposition tree, and the explicit reduction potential. It permits real many-to-one merges and never assumes raw Hall.

The resulting fixed-scope terminal package is determinate and non-degenerate, so kernel governance already holds by necessity. The manuscript-specific population theorem supplies the generated five-way classification of every remaining same-identity collision together with literal certificates for the non-skin branches. Fixed-domain AASC exhaustion removes the attempted independent-authorizer branch; reduction minimality discharges the bounded, split, and sunflower branches; and the resulting four-role closeout excludes two independent standing-bearing occupants at one fixed terminal locus. Generic anchor–tensor–skin classification alone is not asked to establish seed injectivity.

The endpoint bridge then proves that the AASC standing quotient and the independently defined combinatorial endpoint are faithful presentations of the same saturated endpoint. It transfers closeout between presentations but does not generate paths, quotient seed rigidity, coverage rigidity, or raw Hall.

Kernel Principle 1.10 (Integrated-proof declaration). *The combinatorial and AASC components are jointly load-bearing and complete in their assigned roles. The combinatorial component is not a preliminary approximation awaiting an autonomous conventional completion, and the AASC component is not an interpretive supplement to an otherwise independently complete combinatorial argument. The theorem is the composition*

$$\begin{array}{c}
 \text{combinatorial generation, disposition, reduction, terminality, coverage, and transfer} \\
 \Downarrow \\
 \text{AASC impossibility of the forced certificate-terminal countercase} \\
 \Downarrow \\
 \text{source injection, blocker bound, and sunflower endpoint.}
 \end{array}$$

A wholly autonomous conventional replacement of the terminal AASC theorem would constitute a different strengthening, not completion of the theorem proved here.

Kernel Principle 1.11 (Integrated endpoint target). *For every certificate-terminal, fixed-scope, non-degenerate determinate package and exact source fibre U , the proof establishes:*

1. *terminal AASC occupancy rigidity;*
2. *the biconditional $a \equiv_{\text{term}} b \iff a \equiv_{\text{comb}} b$;*
3. *a unique endpoint-faithful equivalence*

$$\mathcal{B}_{G,U} : \text{AASCEnd}_{G,U} \xrightarrow{\sim} \text{CombEnd}_{G,U};$$

4. *injectivity of the terminal seed readout on surviving endpoint classes; and*
5. *injectivity of any selected source-to-seed map after coverage rigidity is applied.*

Scope Statement 1.12 (Raw Hall is neither assumed nor recovered prematurely). The theorem does not assert an injective map from raw roots or raw histories to terminal seeds before disposition. The rank-two triangle refutes that assertion. After terminal disposition and AASC closeout, however, a source-to-seed injection is derived on the surviving exact source fibre. This is a downstream Hall-like cardinal corollary, not a raw residual matching premise in the sense of Hall’s theorem [2].

Scope Statement 1.13 (Coverage-faithful quotienting does not compress the blocker). A quotient may remove duplicate histories over one source or endpoint-neutral presentation multiplicity. For roots of a cardinality-minimal blocker, equality or subsumption of original-edge coverage forces equality of roots. Hence a source representative set that still hits every original edge is the original minimal blocker itself, not a smaller blocker manufactured by quotienting.

Scope Statement 1.14 (The bridge is not a definition by fiat). The AASC relation and the combinatorial relation are independently defined. Their biconditional coincidence is proved after closeout. The bridge expresses simultaneous endpoint identification; it is not a substitute for the terminal exhaustion theorem that precedes it.

Remark 1.15 (Logical restriction, not temporal succession). Standing, Reference, coverage, tensor content, and licensed metric description are jointly instantiated in the constraint-saturated endpoint. Phrases such as “after exhaustion” mark the hypotheses under which a conclusion is authorized. Metricity may be concurrent with Standing while remaining non-authorizing: it reports the population of already fixed endpoint objects and does not create primitive identity [14].

2. Track I: the finite combinatorial carrier

2.1. Core–petal normal form

For $C \subseteq X$ define the link over C by

$$\mathcal{F}_C = \{A \setminus C : A \in \mathcal{F}, C \subseteq A\}.$$

Write $\nu(\mathcal{G})$ for the maximum number of pairwise disjoint members of a finite family $\mathcal{G}$.

A concrete matching of petals is the combinatorial endpoint presentation. The sunflower carrier is the determinate endpoint object represented by that matching; changes of names, path descriptions, residual history, or display do not create additional primitive carriers.

Proposition 2.1 (Core–petal equivalence). *For every $k \geq 2$,*

$$\text{Sun}_k(\mathcal{F}) \iff \exists C \subseteq X (\nu(\mathcal{F}_C) \geq k).$$

In particular,

$$\text{NoSun}_k(\mathcal{F}) \iff \forall C \subseteq X (\nu(\mathcal{F}_C) < k).$$

Proof. If $A_0, \dots, A_{k-1}$ have common pairwise intersection C , then $A_i \setminus C$ are pairwise disjoint members of $\mathcal{F}_C$. Conversely, pairwise disjoint $P_i \in \mathcal{F}_C$ lift to $C \cup P_i \in \mathcal{F}$, and the lifted sets have pairwise intersection exactly C . $\square$

This equivalence fixes the endpoint object. A sunflower is not inferred from a semantic label: it is a concrete matching of petals in a concrete link.

2.2. Carrier object and generated interior domain

The standing carrier and the finite constructor population are distinct types. The carrier is fixed later by the complete admitted probe profile. Track I first proves a positive statement about the objects that actually occur in the residual construction.

Definition 2.2 (Finite actual-constructor domain). *For a fixed fibre U and stage j , let $\mathfrak{G}_{U,j}$ be the finite set of constructor occurrences that actually appear in the records $\Gamma_j(x)$ ($x \in U$), their finite path sets, the fixed certificate ledger, and the finite readout registry generated by Track I. Every occurrence has exactly one of the forms*

$$\begin{aligned} \text{Coord}_j(x) &= \text{a blocker-coordinate occurrence,} \\ \text{Edge}_j(x) &= E_x^{(j)} \text{ attached to } x, \\ \text{Residual}_j(x) &= R_x^{(j)} \text{ attached to } x, \\ \text{Slot}_j(x) &= \text{fibslot}_{R_x^{(j)}}(x), \\ \text{Incidence}_j(x) &= \text{private-edge and blocker-incidence data at } x, \\ \text{Path}_j(\pi) &= \text{an actually generated compatible path,} \\ \text{Bundle}_j(A) &= \text{an actually generated finite composite,} \\ \text{Status}_j(c) &= \text{a literal closeout, split, sunflower, or scope certificate,} \\ \text{Readout}_j(d) &= \text{an actually generated enumeration or display.} \end{aligned}$$

Nested bundle notation is flattened at formation, so a bundle of bundles is identified with the single finite bundle of its constituent occurrences. The fixed scope package D_U is not an interior constructor: it is the anchor package relative to which the finite occurrence domain is evaluated.

Proposition 2.3 (Literal finiteness of the constructor domain). *For every finite U and stage j , the set $\mathfrak{G}_{U,j}$ is finite.*

Proof. Every $\Gamma_j(x)$ contains a finite path set and finitely many finite decorations and certificates. The root set U is finite, and the readout registry is finite. Taking the union of these actual occurrences therefore gives a finite set. Flattening bundle notation prevents an artificial infinite tower of formally nested copies. $\square$

Definition 2.4 (Coordinate support and generated dependency). *Define*

$$\text{csupp}_{U,j} : \mathfrak{G}_{U,j} \longrightarrow \mathcal{P}_{\text{fin}}(B^{(j)})$$

recursively by

$$\begin{aligned} \text{csupp}(\text{Coord}_j(x)) &= \{x\}, \\ \text{csupp}(\text{Edge}_j(x)) &= \{x\}, \\ \text{csupp}(\text{Residual}_j(x)) &= \{x\}, \\ \text{csupp}(\text{Slot}_j(x)) &= \{x\}, \\ \text{csupp}(\text{Incidence}_j(x)) &= \{x\}, \\ \text{csupp}(\text{Path}_j(\pi)) &= \{\text{last}(\pi)\}, \\ \text{csupp}(\text{Bundle}_j(A)) &= \bigcup_{a \in A} \text{csupp}(a), \\ \text{csupp}(\text{Status}_j(c)) &= \text{csupp}(\text{subject}(c)), \\ \text{csupp}(\text{Readout}_j(d)) &= \text{csupp}(\text{subject}(d)). \end{aligned}$$

Write $b \prec_{\text{gen}} a$ when a is formed from, attached to, or displayed from b by one of these constructor clauses.

Theorem 2.5 (Coordinate-root classification). *The $\prec_{\text{gen}}$ -minimal interior constructors in $\mathfrak{G}_{U,j}$ are exactly*

$$\text{Coord}_j(x), \quad x \in B^{(j)}.$$

Every edge, residual, slot, incidence object, path, finite bundle, literal certificate, and readout has a generated coordinate ancestor. In particular, finite composition creates no new primitive dependency root.

Proof. The coordinate constructors have no interior predecessor in the grammar. Every other constructor is introduced with an attached coordinate, a terminal coordinate, a finite family of already generated subjects, or a subject from which it is certified or displayed. Recursion through the finite constructor tree reaches a coordinate. For a finite bundle the support is the union of the supports of its members, so composition cannot create a new root. $\square$

Proposition 2.6 (Generated-domain grammar is non-generative). *Definitions 2.2 and 2.4 introduce no object beyond the finite blocker tower and the outputs of the operations actually used by Track I. They record constructor dependency and do not assert standing, carrier occupancy, seed capacity, or endpoint status.*

Proof. Every clause is an actual coordinate, finite set, local index, path, composite, certificate, or display occurrence extracted from the fixed residual tower. Proposition 2.3 gives literal finiteness. No representative is selected and no governance predicate appears. $\square$

2.3. Maximal matching and the raw blocker

Fix a core C and a maximal pairwise disjoint subfamily $\mathcal{M} \subseteq \mathcal{F}_C$. Put

$$B_{\text{raw}} = \bigcup_{P \in \mathcal{M}} P.$$

Lemma 2.7 (Maximal-matching blocker). *Every nonempty $P \in \mathcal{F}_C$ meets B_{raw} . If $\text{NoSun}_k(\mathcal{F})$ and each link petal has size at most r , then*

$$|\mathcal{M}| \leq k - 1, \quad |B_{\text{raw}}| \leq (k - 1)r.$$

Proof. A petal disjoint from B_{raw} would extend $\mathcal{M}$, contrary to maximality. By Proposition 2.1, $\mathcal{M}$ has fewer than k members. The union bound gives the cardinal estimate. $\square$

At the empty core and rank n , this is the standard rank-dependent Erdős–Rado hitting set. It is a literal blocker, but it does not provide an effective constant-size blocker because its bound grows with n .

2.4. Why the proof is density-restricted

No fixed constant can bound a literal hitting blocker for every three-sunflower-free family at every rank. On the ground set $V \times V$, let

$$S_i = (\{i\} \times V) \cup (V \times \{i\}) \quad (i \in V).$$

If $i \neq j$, then

$$S_i \cap S_j = \{(i, j), (j, i)\}.$$

For three distinct indices these pairwise intersections are different, so the family $\{S_i : i \in V\}$ contains no three-sunflower. Any hitting set $D \subseteq V \times V$ for all pair-stars must nevertheless cover every vertex of V as a first or second coordinate. Hence

$$|V| \leq 2|D|.$$

Taking $|V|$ arbitrarily large defeats every proposed fixed all-family literal blocker bound. This does not refute the sunflower lemma; it refutes the stronger and unnecessary assertion that a fixed literal blocker is available before the quantitative counterexample is imposed.

Accordingly, all later population sources are requested only in a *dense counterexample*:

$$|\mathcal{F}| > B^n.$$

If this inequality fails, the desired endpoint estimate already holds. This restriction is exact and prevents a false all-family source from entering the proof.

Audit Gate 2.8 (Star-family regression). *A step asserting that every sunflower-free family has a literal hitting blocker of size at most B is invalid. The permitted conclusion is that every dense endpoint counterexample has an effective blocker of size at most B , and that conclusion is obtained only after the support-fibre closure.*

2.5. Canonical support cells and the conservative ambient alphabet

Fix the empty-core maximal matching used to form the canonical raw blocker. Its petals are pairwise disjoint. Consequently, every coordinate of B_{raw} belongs to exactly one matching petal. After embedding the at most $k - 1$ petals into $\{0, \dots, k - 1\}$, encode that unique incidence as a singleton subset. The formal carrier therefore uses

$$\text{supp} : B_{\text{min}} \longrightarrow \mathcal{P}(\{0, 1, 2\})$$

for $k = 3$.

Lemma 2.9 (Singleton support-cell realization). *For every $x \in B_{\text{min}}$, the set $\text{supp}(x)$ has cardinality one. Thus only singleton cells are realized, although the formal codomain is the full eight-element powerset $\mathcal{P}(\{0, 1, 2\})$.*

Proof. The minimal blocker is contained in the raw blocker. A raw-blocker coordinate lies in at least one matching petal by construction and in at most one because the matching petals are pairwise disjoint. Its encoded support is therefore a singleton. $\square$

The full powerset is retained as the uniform ambient support alphabet used by the Lean interface. Counting all eight ambient cells is conservative: fibres over unrealized cells are empty. The local theorem bounds each fibre $\text{supp}^{-1}(S)$, and the global blocker bound sums over the complete ambient alphabet. No quotient representative replaces the literal blocker, and no live edge hit is discarded.

3. Minimal blockers, private witnesses, and the branch-complete generator

3.1. Cardinality-minimal blocker

Let B_{raw} be the finite blocker from the empty-core maximal matching. Choose a hitting subset $B \subseteq B_{\text{raw}}$ of minimum cardinality and write $B_{\text{min}} = B$. The choice is made inside a finite nonempty collection; no representative-selection rule receives standing-bearing force.

Lemma 3.1 (Private edge). *For every $x \in B$ there is an edge $E_x \in \mathcal{F}$ satisfying*

$$E_x \cap B = \{x\}.$$

Moreover, $x \mapsto E_x$ is injective.

Proof. If no such edge existed, $B \setminus \{x\}$ would still hit every edge, contradicting cardinality minimality. If $E_x = E_y$, then its intersection with B is both $\{x\}$ and $\{y\}$, hence $x = y$. $\square$

Definition 3.2 (Private-witness endpoint load). *For $x \neq y$ in B_{min} , the private-witness endpoint load $\text{PWLoad}_D(x, y)$ is the incidence record*

$$E_x, E_y; \quad x \in E_x, \quad y \notin E_x, \quad y \in E_y, \quad x \notin E_y.$$

Lemma 3.3 (Distinct private witnesses carry a load). *For $x, y \in B_{\text{min}}$,*

$$x \neq y \implies \text{PWLoad}_D(x, y).$$

Proof. The positive incidences are part of the private-edge construction. Since $E_x \cap B_{\text{min}} = \{x\}$ and $y \in B_{\text{min}} \setminus \{x\}$, one has $y \notin E_x$; the other negative incidence is symmetric. $\square$

3.2. Skin collapse is literal on the blocker carrier

Define endpoint-incidence equivalence by

$$x \sim_{\text{inc}} y \iff \forall E \in \mathcal{F} (x \in E \iff y \in E).$$

Proposition 3.4 (Private-witness finality). *For $x, y \in B_{\min}$,*

$$x \sim_{\text{inc}} y \iff x = y.$$

Consequently, endpoint skin-equivalence on the literal minimal blocker is identity.

Proof. Evaluate incidence on E_x . Since $x \in E_x$ and $E_x \cap B_{\min} = \{x\}$, equivalent incidence forces $y = x$. The reverse implication is immediate. $\square$

3.3. Residual family and local parent slots

For $x \in B_{\min}$ put

$$R_x = E_x \setminus \{x\},$$

and let

$$\mathcal{R}(\mathcal{F}) = \{R_x : x \in B_{\min}\}.$$

This is an $(n-1)$ -uniform family. Equal residuals may have several parents, but their multiplicity is bounded.

Lemma 3.5 (Residual-fibre bound). *If $\mathcal{F}$ contains no k -sunflower, then for every residual R ,*

$$|\{x \in B_{\min} : R_x = R\}| < k.$$

For $k = 3$, every residual has at most two parents.

Proof. If distinct $x_0, \dots, x_{k-1}$ had the same residual R , then $E_{x_i} = R \cup \{x_i\}$. The x_i are distinct and absent from R , so the private edges have common pairwise intersection R and form a k -sunflower. $\square$

Fix once and for all a finite enumeration of each residual fibre. Write

$$\text{fibslot}_{R_x}(x) \in \text{Fin}(3)$$

for the resulting local position. This is generated combinatorial data; it is not the later 4094-place seed readout.

Proposition 3.6 (Residual-plus-slot injectivity). *The map*

$$x \longmapsto (R_x, \text{fibslot}_{R_x}(x))$$

is injective on $B_{\min}$.

Proof. Equal first coordinates place the two parents in one residual fibre. Equal local slots then identify the same enumerated parent. $\square$

Definition 3.7 (Native residual distinction). *NativeDist $_D(x, y)$ means that the difference between x and y is witnessed by private-edge incidence, unequal residuals, unequal local residual slots, a point of $R_x \triangle R_y$, or a declared continuation record.*

Proposition 3.8 (Native content is not independent authority). *A native residual distinction is lawful combinatorial content. By itself it does not create a second target, admissibility gate, primitive carrier source, or endpoint authorizer.*

Proof. The distinction is generated finite data. Endpoint authority is a fixed- domain use role affecting admissibility or theorem status. Raw difference does not supply such authority without a standing-bearing role certificate. $\square$

3.4. Raw private-witness identity and provenance

The inherited object must be typed before it is displayed numerically.

Definition 3.9 (Raw private-witness identity). *For $x \in B_{\min}$ define*

$$\text{Rawld}_D(x) = (x, E_x, R_x, \text{fibslot}_{R_x}(x), \text{supp}(x)).$$

The support field is the canonical matching cell from Section 2. The explicit profile and role fields are appended in Section 4 after their predicates are defined; they do not alter the generator.

Lemma 3.10 (Raw identity is injective). *The map $x \mapsto \text{Rawld}_D(x)$ is injective.*

Proof. It contains the injective residual-plus-slot code of Proposition 3.6; it also retains the private edge and source coordinate. $\square$

Definition 3.11 (Source-root datum). *For $x \in U \subseteq B^{(0)}$, define the complete stage-zero source datum*

$$\text{Root}_U(x) = (x, E_x^{(0)}, R_x^{(0)}, \text{fibslot}_{R_x^{(0)}}(x), \text{Inc}_0(x)).$$

It is formed before residual continuation, seed readout, or governance classification.

Proposition 3.12 (Private-witness root determinacy). *For $x, y \in U$,*

$$\text{Root}_U(x) = \text{Root}_U(y) \iff x = y.$$

In particular, $x \neq y$ implies $\text{Inc}_0(x) \neq \text{Inc}_0(y)$.

Proof. If $x \neq y$, the private edge $E_x^{(0)}$ contains x and omits y , because $E_x^{(0)} \cap B^{(0)} = \{x\}$. Evaluating the incidence signatures on $E_x^{(0)}$ separates the two roots. The converse is immediate. $\square$

Proposition 3.13 (Source-root non-degeneracy). *Every $x \in U$ is a finite determinate, non-degenerate theorem object. It has a fixed reference, is separated from every other source root by private-edge incidence, and is used load-bearingly as one member of the population whose cardinality is bounded. Deleting or merging x changes the submitted source population and the endpoint claim.*

Proof. Finiteness and fixed reference follow from Definition 3.11; separation is Proposition 3.12. The fibre cardinality is the number of these distinct source objects. Hence the proof cannot count their individual identities while treating those identities as standing-neutral bookkeeping. $\square$

Definition 3.14 (Reconstructible private-witness provenance). *The stage-zero provenance trace of x is*

$$\text{Prov}_D(x) = (E_x, R_x, \text{fibslot}_{R_x}(x)).$$

At later ranks the generated provenance is the complete finite rooted lineage of private edges, residuals, local fibre positions, and blocker incidences defined below. A provenance field is tensor content when it changes an admitted probe value and skin when it is neutral under the complete probe family. In neither case is it a second primitive carrier or an independent source of standing.

Proposition 3.15 (Provenance is a descriptive asymmetry marker). *Private-witness provenance is generated before continuation use, stable under lawful fixed-carrier redescription, and independent of later success. It may distinguish generated records and support exact reconstruction, but it has no primitive authority. No environmental boundary trace or canonical representative is required to retain it.*

Proof. The private edge and local residual position are determined from the submitted finite family and minimal blocker before any endpoint collision is tested. They are not introduced after failure and are preserved by every same-family redescription that preserves the fixed carrier and private-parent interface. The role-fixation normal forms of [7, 6] list internal invariants and declared provenance markers before the boundary-trace route. The present trace occupies those descriptive branches without acquiring a second primitive referent. $\square$

Remark 3.16 (Information is not primitive authority). Distinct provenance traces may carry different information and may be needed for truncation or audit. Such differentiation does not multiply primitive carriers unless fixed Reference independently establishes different referents.

3.5. Sunflower-free residual tower

Set $\mathcal{F}^{(0)} = \mathcal{F}$. Whenever $\mathcal{F}^{(j)}$ has positive rank, fix a cardinality-minimal blocker $B^{(j)}$, a private edge $E_y^{(j)}$ for each $y \in B^{(j)}$, and a finite enumeration of every residual-parent fibre. Put

$$R_y^{(j)} = E_y^{(j)} \setminus \{y\}, \quad \mathcal{F}^{(j+1)} = \{R_y^{(j)} : y \in B^{(j)}\}.$$

All tower choices are fixed before the crossing and belong to the endpoint package D .

Lemma 3.17 (Residual sunflower lifting). *If $\mathcal{F}^{(j+1)}$ contains a three-sunflower, then $\mathcal{F}^{(j)}$ contains a three-sunflower. Hence*

$$\text{NoSun}_3(\mathcal{F}^{(0)}) \implies \text{NoSun}_3(\mathcal{F}^{(j)})$$

for every stage of the tower.

Proof. Choose the three residual petals and one private parent for each. A private coordinate belongs to no other private residual because every private edge meets $B^{(j)}$ in its own coordinate only. Reinserting the three distinct private coordinates therefore preserves the common residual core and lifts the three residuals to a three-sunflower in $\mathcal{F}^{(j)}$. $\square$

At stage j , $\mathcal{F}^{(j)}$ is $(n - j)$ -uniform. Thus rank is a strictly decreasing termination measure.

Lemma 3.18 (Strict tower termination). *Let $N = 2047$ and $m = \max(0, n - N)$. The residual tower reaches the family $\mathcal{F}^{(m)}$ of rank at most N after exactly m steps, and no stage can reopen a higher rank.*

Proof. Residual formation lowers the uniform rank by one. Integer rank is therefore a well-founded measure. $\square$

3.6. Branch-complete sourcewise continuation

At every stage j , fix a cardinality-minimal blocker $B^{(j)}$, choose one private edge $E_x^{(j)}$ for each $x \in B^{(j)}$, and enumerate every residual fibre. Put

$$R_x^{(j)} = E_x^{(j)} \setminus \{x\}, \quad \lambda_j(x) = (E_x^{(j)}, R_x^{(j)}, \text{fibslot}_{R_x^{(j)}}(x)).$$

All choices are made from finite nonempty sets before any continuation status is evaluated.

Definition 3.19 (Decorated coordinate presentation candidate). *For $x \in B^{(j)}$, define its full stage- j incidence signature by*

$$\text{Inc}_j(x) = (\mathbf{1}_{x \in E})_{E \in \mathcal{F}^{(j)}}$$

and its decorated coordinate by

$$\widehat{x}^{(j)} = (x, E_x^{(j)}, R_x^{(j)}, \text{fibslot}_{R_x^{(j)}}(x), \text{Inc}_j(x)).$$

Let

$$\widehat{B}^{(j)} = \{\widehat{x}^{(j)} : x \in B^{(j)}\}.$$

This is a generated coordinate-presentation population. No standing or carrier claim is part of the definition.

Proposition 3.20 (Primitive coordinate-presentation generation). *The projection*

$$\widehat{B}^{(j)} \longrightarrow B^{(j)}, \quad \widehat{x}^{(j)} \longmapsto x,$$

is bijective. Distinct decorated coordinates have distinct private-edge incidence signatures.

Proof. Surjectivity and injectivity of the projection are immediate from the construction. If $x \neq y$, the private edge $E_x^{(j)}$ contains x and omits y , so $\text{Inc}_j(x) \neq \text{Inc}_j(y)$. $\square$

Definition 3.21 (Next-incidence fibre). *For $0 \leq j < m$ and $x \in B^{(j)}$, define*

$$\mathcal{I}_j(x) = R_x^{(j)} \cap B^{(j+1)}.$$

This is a set of blocker coordinates, not an identification of the residual edge with a blocker point.

Lemma 3.22 (Next-incidence nonemptiness). *For every $0 \leq j < m$ and $x \in B^{(j)}$,*

$$\mathcal{I}_j(x) \neq \emptyset.$$

Proof. The residual $R_x^{(j)}$ is an edge of $\mathcal{F}^{(j+1)}$. The set $B^{(j+1)}$ is a hitting blocker for every edge of $\mathcal{F}^{(j+1)}$. Hence $R_x^{(j)} \cap B^{(j+1)}$ is nonempty. $\square$

Definition 3.23 (Compatible decorated path). *Let $U \subseteq B^{(0)}$. A compatible path of depth j rooted at $x \in U$ is a tuple*

$$\pi = (x_0, x_1, \dots, x_j)$$

with $x_0 = x$, $x_i \in B^{(i)}$, and

$$x_{i+1} \in \mathcal{I}_i(x_i) \quad (0 \leq i < j).$$

Its decoration is

$$\text{Dec}_j(\pi) = (\lambda_0(x_0), \lambda_1(x_1), \dots, \lambda_j(x_j)).$$

Write $\text{Path}_j(x)$ for the finite set of all decorated compatible paths of depth j rooted at x .

Lemma 3.24 (Finite nonempty path population). *For every $x \in U$ and $0 \leq j \leq m$, the set $\text{Path}_j(x)$ is finite and nonempty.*

Proof. At depth zero it contains the one path (x) . If $\text{Path}_j(x)$ is finite and nonempty, extend each path π through every point of the finite nonempty set $\mathcal{I}_j(\text{last}(\pi))$. The union of these finite nonempty extension fibres is finite and nonempty. Induction gives the result. $\square$

Definition 3.25 (Branch-complete continuation record). *For $x \in U$, define*

$$\Gamma_j(x) = (x, \text{supp}(x), \text{prof}(x), \text{role}(x), \text{Path}_j(x)),$$

where every path is stored with its full decoration. A generated closeout, split, sunflower, target-change, nonstanding, or authorizer certificate attached to any path prefix is retained as a finite labelled field of the record. No branch is selected or discarded.

Set

$$\text{Raw}_j(U) = \{\Gamma_j(x) : x \in U\}.$$

Definition 3.26 (Typed current-stage record). *Let $\pi \in \text{Path}_j(x)$ and put $y = \text{last}(\pi)$. The corresponding current-stage record is typed as*

$$\text{Rec}_j(x, \pi) = (\text{trace}_j(x), y, \Theta_j(x, \pi), \text{skin}_j(x, \pi)).$$

The four components have distinct roles:

- $\text{trace}_j(x)$ is the retained historical lineage marker;
- $y \in B^{(j)}$ is the current coordinate-presentation constructor;
- $\Theta_j(x, \pi)$ is the endpoint-relevant tensor profile, including private-edge, residual, local-slot, incidence, and continuation data; and
- the final skin component contains only neutral ordering, enumeration, notation, and display choices.

This typing distinguishes information from primitive authority. The trace may be retained for reconstruction, and the tensor profile may remain consequence-bearing, but neither is a second primitive carrier constructor.

Proposition 3.27 (Stage-zero identity and later trace occurrence). *At stage zero the source coordinate x is both the current coordinate and the historical lineage origin. For every $j > 0$, the retained occurrence of the symbol x in $\Gamma_j(x)$ is a field of $\text{trace}_j(x)$; it is not an additional current coordinate in $B^{(j)}$ and does not create a second primitive presentation locus.*

Proof. At $j = 0$ the unique path is (x) and its last coordinate is x . At positive rank, every current coordinate is generated by the next-incidence recursion and lies in $B^{(j)}$, whereas the source field is copied unchanged by complete extension. The two occurrences therefore have different constructor types: one is the current blocker-coordinate subject and the other is the retained lineage marker. $\square$

Proposition 3.28 (Root reconstruction). *For each j , the map*

$$\gamma_j : U \longrightarrow \text{Raw}_j(U), \quad x \longmapsto \Gamma_j(x),$$

is injective.

Proof. The root coordinate is a retained field of $\Gamma_j(x)$. Equivalently, the depth-zero prefix contains $\text{Rawld}_D(x)$, whose residual-plus-slot component is injective by Proposition 3.6. $\square$

Definition 3.29 (Historical root-trace projection). *Define*

$$\text{hist}_j : \text{Raw}_j(U) \longrightarrow U, \quad \text{hist}_j(\Gamma_j(x)) = x.$$

The map is well defined by Proposition 3.28. For $j > 0$ it is a historical and cross-stage audit map, not the active current- reference assignment on $B^{(j)}$.

Definition 3.30 (Complete extension). *For a path $\pi \in \text{Path}_j(x)$, define its complete one-step extension fibre by*

$$\text{Child}_j(\pi) = \{(\pi, z) : z \in \mathcal{I}_j(\text{last}(\pi))\}.$$

The complete record extension is

$$\text{Ext}_j(\Gamma_j(x)) = \Gamma_{j+1}(x),$$

obtained by replacing every leaf by all members of its child fibre and adding the corresponding terminal decorations and generated certificates.

Definition 3.31 (Prefix truncation). *For $b = \Gamma_{j+1}(x)$, define $\text{Trunc}_j(b)$ by deleting the terminal coordinate and terminal decoration from every depth- $(j+1)$ path, deleting terminal-only certificate fields, and taking the resulting finite set of prefixes. Root and fixed coarse fields are unchanged.*

Theorem 3.32 (Historical trace preservation). *For every $0 \leq j < m$,*

$$\text{hist}_{j+1} \circ \text{Ext}_j = \text{hist}_j.$$

On the image of Ext_j one also has

$$\text{hist}_j \circ \text{Trunc}_j = \text{hist}_{j+1}.$$

Thus complete extension changes the current generated description but not the historical lineage marker. Active current Reference is separately governed on the current coordinate locus.

Proof. By definition $\text{Ext}_j(\Gamma_j(x)) = \Gamma_{j+1}(x)$ and truncation returns $\Gamma_j(x)$. Both records retain the same historical trace field x . $\square$

Theorem 3.33 (Generated extension–truncation retraction). *For every $0 \leq j < m$,*

$$\text{Trunc}_j \circ \text{Ext}_j = \text{id}_{\text{Raw}_j(U)}.$$

Consequently, Ext_j is injective.

Proof. Every old path has at least one child by Lemma 3.22, so it occurs as the prefix of an extended path. Every extended path was formed from one old path, so truncation produces no new prefix. The decorations and prefix certificate fields are copied unchanged. Hence the truncated extension is exactly the original record. A map with a left inverse is injective. $\square$

Theorem 3.34 (Branch-complete combinatorial generation). *For every finite root set $U \subseteq B^{(0)}$, every $x \in U$, and every $0 \leq j \leq m$:*

1. $\Gamma_j(x)$ is a finite nonempty record;
2. every compatible sequence of residual/blocker incidences occurs in it;
3. no path occurs unless every transition satisfies $x_{i+1} \in R_{x_i}^{(i)} \cap B^{(i+1)}$;
4. $\Gamma_{j+1}(x) = \text{Ext}_j(\Gamma_j(x))$;
5. prefix truncation reconstructs $\Gamma_j(x)$; and
6. the construction uses no standing predicate, seed capacity, finite slot, fibre bound, or endpoint theorem.

Proof. Items 1–3 are Definitions 3.23–3.25 and Lemma 3.24. Item 4 is the definition of complete extension, and item 5 is Theorem 3.33. The recursive construction uses only the residual tower, blockers, private edges, local fibre enumerations, and finite certificate verification. $\square$

Theorem 3.35 (Generated-domain saturation). *For every fixed fibre U , stage j , and mathematical object a introduced, retained, transformed, compared, or evaluated by the residual construction in D_U , either a is a fixed anchor of D_U or*

$$a \in \mathfrak{G}_{U,j}.$$

Moreover:

1. *every non-anchor, non-status interior object has nonempty coordinate support;*
2. *every non-coordinate constructor is dependent data over its coordinate support;*
3. *finite composition creates no new dependency root;*
4. *complete extension maps saturated generated domains to saturated generated domains;*
5. *prefix truncation preserves the generated dependency structure; and*
6. *no unlisted same-scope object is produced by an operation in Track I.*

Proof. Use structural induction over the operations of the generator. At stage zero, the private-witness operation produces a coordinate, its private edge, residual, local slot, and incidence data, all listed in Definition 2.2 and rooted at that coordinate. A compatible-path step appends a coordinate from the next-incidence fibre and its listed attachments. Forming the complete path population and the record $\Gamma_j(x)$ is finite bundle formation, whose support is the union of the supports of its members. Literal closeout, split, sunflower, and scope data enter only through the status constructor; enumeration and numerical display enter only through the readout constructor. These are all operations used in Track I. The coordinate-root and composite claims follow from Theorem 2.5.

Complete extension repeats the same listed path-extension operation at the next stage, so it preserves saturation. Prefix truncation deletes only the terminal coordinate layer and terminal-only attachments; all retained prefixes remain generated by the earlier-stage clauses. Hence no operation introduces an unlisted same-scope object. $\square$

Corollary 3.36 (No rootless interior constructor). *If $a \in \mathfrak{G}_{U,j}$ is neither a fixed anchor nor a literal discharge or scope certificate, then*

$$\text{csupp}_{U,j}(a) \neq \emptyset.$$

Corollary 3.37 (Composite non-emergence). *For every finite generated composite A ,*

$$\text{csupp}_{U,j}(\text{Bundle}_j(A)) = \bigcup_{a \in A} \text{csupp}_{U,j}(a).$$

Thus a path tree, relation, provenance bundle, or complete generated record may carry composite tensor content, but it does not become an additional primitive dependency root.

Theorem 3.38 (Coordinate-presentation type preservation). *Complete extension preserves the generated presentation type. More precisely, every child appended to a depth- j path is a decorated coordinate in $\widehat{B}^{(j+1)}$, and every decorated coordinate occurring at stage $j+1$ occurs only as such a child. Complete extension may add incidence, residual, provenance, or status data, but it does not promote a path set, provenance ledger, relation, or numerical display into a coordinate presentation.*

Proof. By Definition 3.30, each child has the form $z \in R_x^{(j)} \cap B^{(j+1)}$. Definition 3.19 then supplies the unique corresponding element $\hat{z}^{(j+1)}$. Conversely, the generator records presentation candidates only through blocker coordinates and their fixed decorations. The remaining fields are attached data in the grammar of Definition 2.2. $\square$

Remark 3.39 (Complete path bundles are generated descriptive populations). The branch-complete tree is indispensable for generation, reconstruction, and probe evaluation. It is a finite population of coordinate occurrences and attached tensor or skin data, not a proof of raw coordinate injectivity. The terminal standing quotient is formed from the complete record, so any continuation- or coverage-bearing composite content remains tensor. Only representative multiplicity below that complete tensor profile is collapsed.

3.7. Strict generated-record quotient

Definition 3.40 (Generated-record equivalence). *For $a, b \in \text{Raw}_j(U)$, write $a \sim_{\text{gen},j} b$ when there is an isomorphism of their finite rooted prefix trees that preserves:*

1. *the root private-witness reference and fixed support/profile/role fields;*
2. *every coordinate constructor, private edge, residual, local residual slot, and incidence relation;*
3. *every generated obstruction, closeout, split, and endpoint-status field;*
4. *the complete provenance trace;*
5. *prefix and child relations, including the extension/truncation interface.*

The isomorphism may change only representative names, finite enumeration order, chart labels, route notation, and equivalent displays. It contains no standing or carrier classification.

Proposition 3.41 (Generated-record equivalence is an equivalence relation). *The relation $\sim_{\text{gen},j}$ is reflexive, symmetric, and transitive.*

Proof. Use the identity tree isomorphism, inverse isomorphism, and composition of structure-preserving isomorphisms. $\square$

Definition 3.42 (Strict generated quotient). *Define*

$$\text{QGen}_j(U) = \text{Raw}_j(U) / \sim_{\text{gen},j}.$$

No standing predicate, probe relation, or survivor restriction occurs in this quotient. Section 5 defines the live population and the canonical standing quotient induced by the complete admitted probe family.

Remark 3.43 (Two quotient levels). The strict quotient preserves every generated combinatorial and provenance field needed by truncation. The later standing quotient identifies generated descriptions exactly when every admitted probe agrees. Probe-detected incidence and continuation differences remain tensor content; only variation below the fixed tensor profile is skin.

Proposition 3.44 (Extension and truncation descend to the generated quotient). *The maps Ext_j and Trunc_j respect generated-record equivalence and induce*

$$\overline{\text{Ext}}_j : \text{QGen}_j(U) \longrightarrow \text{QGen}_{j+1}(U), \quad \overline{\text{Trunc}}_j : \text{QGen}_{j+1}(U) \longrightarrow \text{QGen}_j(U),$$

with

$$\overline{\text{Trunc}}_j \circ \overline{\text{Ext}}_j = \text{id}_{\text{QGen}_j(U)}.$$

Proof. A structure-preserving isomorphism extends leafwise through the corresponding incidence fibres and truncates prefixwise. Hence equivalent records have equivalent extensions and truncations. Apply Theorem 3.33 to representatives. $\square$

Proposition 3.45 (Coordinate support descends to the generated quotient). *Generated-record equivalence preserves coordinate support. Hence*

$$\overline{\text{csupp}}_{U,j} : \mathbf{QGen}_j(U) \longrightarrow \mathcal{P}_{\text{fin}}(B^{(j)})$$

is well defined. Every class represented by a branch-complete record has nonempty support.

Proof. Definition 3.40 preserves every coordinate constructor, incidence relation, and prefix/child relation. It therefore preserves the terminal coordinates and the recursive support of every attached constructor. Nonemptiness follows from Theorem 3.35 and Lemma 3.24. $\square$

Proposition 3.46 (Initial generated-quotient injection). *The map*

$$\eta_0^{\text{gen}} : U \hookrightarrow \mathbf{QGen}_0(U), \quad x \longmapsto [\Gamma_0(x)]_{\text{gen}},$$

is injective.

Proof. Generated-record equivalence preserves the root private-witness reference and its private-edge, residual, local-slot, and provenance data. Proposition 3.28 therefore descends to the quotient. $\square$

Audit point. Track I has positively generated the complete sourcewise population, proved the extension–truncation retraction, and established generated-domain saturation with nonempty coordinate support. Section 5 forms the terminal standing and coverage quotient of this already generated population. It does not generate a successor, coordinate, matching, or raw Hall selector.

4. Combinatorial seed capacity and the exact high-rank frontier

4.1. Controlled seed population

The classical Erdős–Rado recurrence gives a factorial-type upper bound for fixed k [1]. In the mechanized three-petal specialization, the verified cutoff is

$$N = 2047.$$

Every minimal blocker at rank at most N has cardinality at most

$$(3 - 1)N = 4094.$$

Theorem 4.1 (Seed population). *Let $\mathcal{G}$ be an s -uniform three-sunflower-free family with $s \leq 2047$. There is an injective map*

$$\text{seedCode}_{\mathcal{G}} : B_{\text{min}}(\mathcal{G}) \hookrightarrow \text{Fin}(4094).$$

Proof. The maximal-matching raw blocker has cardinality at most $2s \leq 4094$. A cardinality-minimal blocker is a subset of it and therefore embeds in $\text{Fin}(4094)$. This is the traditional seed embedding verified in the Lean companion. $\square$

Remark 4.2 (Capacity, not universal semantics). The map $\text{seedCode}_{\mathcal{G}}$ is family-local. Its values are numerical readouts of already identified seed presentations. No theorem requires the same numeral in two different seed families or fibre scopes to denote one common standing state.

This ordering follows the residue/readout discipline of [9]: an object is first typed in its carrier or role fibre; coordinate projection is then a non-generative display of that object.

4.2. Seed incidence tensors and decorated points

Fix, for each $z \in B_{\min}(\mathcal{G})$, one private edge $E_z^{\mathcal{G}}$ satisfying $E_z^{\mathcal{G}} \cap B_{\min}(\mathcal{G}) = \{z\}$. Define the full seed incidence signature

$$\text{Inc}_{\mathcal{G}}(z) = (\mathbf{1}_{z \in E})_{E \in \mathcal{G}}$$

and the decorated seed point

$$\widehat{z} = (z, E_z^{\mathcal{G}}, \text{Inc}_{\mathcal{G}}(z)).$$

Put

$$\widehat{B}_{\mathcal{G}} = \{\widehat{z} : z \in B_{\min}(\mathcal{G})\}.$$

The coordinate z is the literal presentation component; its private edge and incidence signature are separating tensor data. Section 5 assigns each coordinate its terminal standing-reference status inside the fixed fibre. That status is governance typing of this already generated point, not an additional coordinate or numerical slot, so the decorated population remains in bijection with $B_{\min}(\mathcal{G})$.

Proposition 4.3 (Decorated seed-point equivalence). *The projection*

$$\pi_{\mathcal{G}} : \widehat{B}_{\mathcal{G}} \longrightarrow B_{\min}(\mathcal{G}), \quad \widehat{z} \longmapsto z,$$

is a bijection. Consequently,

$$|\widehat{B}_{\mathcal{G}}| = |B_{\min}(\mathcal{G})| \leq 4094.$$

Proof. The definition gives exactly one decorated point for each blocker coordinate. The cardinal estimate is Theorem 4.1. $\square$

Lemma 4.4 (Seed tensor separation). *For $z_1, z_2 \in B_{\min}(\mathcal{G})$,*

$$\text{Inc}_{\mathcal{G}}(z_1) = \text{Inc}_{\mathcal{G}}(z_2) \iff z_1 = z_2.$$

Equivalently, distinct decorated seed points have distinct private-edge incidence tensors.

Proof. If the incidence signatures agree, evaluate them on the private edge $E_{z_1}^{\mathcal{G}}$. It contains z_1 and meets the minimal blocker only at z_1 , so it cannot contain a distinct z_2 . The reverse implication is immediate. $\square$

4.3. Generated finite type coordinates

For $x \in B^{(0)}$, let

$$d_{\mathcal{F}}(x) = |\{E \in \mathcal{F} : x \in E\}|, \quad \mu(x) = |\{y \in B^{(0)} : R_y^{(0)} = R_x^{(0)}\}|.$$

By Lemma 3.5, $\mu(x) \in \{1, 2\}$ in a three-sunflower-free package. Define six bivalent observations:

$$\begin{aligned} \circ_1(x) &\iff \mu(x) = 2, \\ \circ_2(x) &\iff d_{\mathcal{F}}(x) = 1, \\ \circ_3(x) &\iff R_x^{(0)} \cap B_{\text{raw}} \neq \emptyset, \\ \circ_4(x) &\iff \exists y \in B^{(0)} \setminus \{x\} \ R_x^{(0)} \cap R_y^{(0)} \neq \emptyset, \\ \circ_5(x) &\iff \exists E \in \mathcal{F} \setminus \{E_x^{(0)}\} \ R_x^{(0)} \subseteq E, \\ \circ_6(x) &\iff |R_x^{(0)}| > 0 \ \wedge \ \exists y \in B^{(0)} \setminus \{x\} \ |R_x^{(0)} \cap R_y^{(0)}| + 1 = |R_x^{(0)}|. \end{aligned}$$

Set

$$\text{prof}(x) = \{i \in \{1, \dots, 6\} : \circ_i(x)\}.$$

Each observation is a finite decidable predicate on generated combinatorial data; hence prof is a total map with at most 2^6 values. They are used only as explicit finite type coordinates in the global code; no claim is made that they exhaust all tensor information or independently bear primitive standing.

4.4. Explicit four-way priority role partition

Define the branching predicate

$$\text{Br}_0(x) \iff m > 0 \ \wedge \ |\mathcal{I}_0(x)| \geq 2.$$

Define $\text{role} : B^{(0)} \rightarrow \{\text{Skin}, \text{BoundedFiber}, \text{TensorSplit}, \text{SunRole}\}$ by the priority rule

$$\text{role}(x) = \begin{cases} \text{SunRole}, & \text{if a generated concrete three-sunflower certificate contains } x, \\ \text{TensorSplit}, & \text{otherwise, if } \text{Br}_0(x), \\ \text{BoundedFiber}, & \text{otherwise, if } \mu(x) = 2, \\ \text{Skin}, & \text{otherwise.} \end{cases}$$

The priority order makes the values mutually exclusive and exhaustive. These four names are coarse finite tags only. They are not ATS role assignments and do not themselves assert standing, splitting, or a sunflower.

For a support cell S , profile p , and role r , put

$$U_{S,p,r} = \{x \in B^{(0)} : \text{supp}(x) = S, \text{prof}(x) = p, \text{role}(x) = r\}.$$

The profile/role signature has at most $2^6 \cdot 4 = 256$ values.

Theorem 4.5 (Generated typed-coordinate certificate). *The support cell, six profile bits, and four-way priority role are total finite functions generated from Track-I data before seed readout. Their values are:*

1. $\text{supp}(x)$, the canonical matching-petal type;
2. $\circ_1$, residual-parent multiplicity one versus two;
3. $\circ_2$, degree one versus repeated family incidence;
4. $\circ_3$, contact with the raw blocker;

5. $\circ_4$, residual overlap with another source root;
6. $\circ_5$, residual containment in another family edge;
7. $\circ_6$, near-total residual overlap; and
8. role, the first applicable generated branch tag under the fixed priority rule.

Consequently

$$\Sigma(x) = (\text{supp}(x), \text{prof}(x), \text{role}(x))$$

is defined independently of the seed coordinate, seed code, fibre bound, and endpoint estimate, and takes at most

$$8 \cdot 2^6 \cdot 4 = 2048$$

values. This theorem asserts generated finite typing, not that every type coordinate is an additional primitive standing source.

Proof. Every entry is evaluated by an explicit predicate on the fixed matching, blocker, private-edge, residual, degree, containment, overlap, or next-incidence data. Each codomain is finite and every value is fixed before the controlled seed is enumerated. No entry depends on the desired estimate. $\square$

Theorem 4.6 (Typed endpoint-code architecture). *Let*

$$\mathcal{T} = \text{im}(\text{supp}) \times \{0, 1\}^6 \times \{\text{Skin}, \text{BoundedFiber}, \text{TensorSplit}, \text{SunRole}\}.$$

Then $|\mathcal{T}| \leq 2048$. For each exact source fibre $U_{S,p,r}$, a surviving endpoint class is read by

$$(S, p, r, z) \in \mathcal{T} \times B_{\min}(G), \quad z = \bar{\epsilon}_{G, U_{S,p,r}}(C).$$

Once certificate-terminal closeout has made the seed readout injective on endpoint classes, the assembled endpoint map

$$C \mapsto (S, p, r, \bar{\epsilon}_{G, U_{S,p,r}}(C))$$

is injective on the disjoint union of the endpoint spaces. No injection of raw roots into seeds is assumed at this stage. Section 6 derives source injectivity only after terminal closeout and minimal-blocker coverage rigidity.

Proof. Equal global codes give equal support, profile, role, and terminal seed. The first three coordinates place the classes in one exact fibre; quotient-final seed rigidity then gives equality of endpoint classes. The codomain bound follows from the displayed finite factors. The statement concerns saturated endpoint classes and makes no assertion about the pre-discharge raw merge relation. $\square$

Theorem 4.7 (No strengthening by post hoc refinement). *Let λ be an additional finite label introduced after the typed quotient code has been fixed. Appending λ produces a codomain with an additional factor $|\text{im } \lambda|$. If λ changes no terminal standing, coverage, continuation, or reconstruction probe, it is skin and cannot create another carrier class. If it does change one of those probes, it is tensor content and must be included explicitly in the quotient profile. In neither case can an uncounted label multiply the proved capacity. This is the finite-sector refinement discipline used here [10].*

Proof. A standing-neutral label factors through the terminal standing quotient and leaves the class unchanged. A standing-relevant label distinguishes quotient classes and therefore belongs to the tensor profile used to define them. Ordinary typed counting then charges every displayed coordinate exactly once. $\square$

Remark 4.8 (Profile and role are paid-for type coordinates). The values S, p, r are generated before readout and type the global code. They are not asserted to be primitive carriers or to exhaust all tensor information. Full coverage and continuation data remain in the terminal standing quotient rather than being compressed into the six-bit profile.

The finite type count is

$$|\mathcal{T}| \leq 8 \cdot 2^6 \cdot 4 = 2048.$$

Section 5 proves terminal seed rigidity on the surviving endpoint classes. Section 6 then combines that result with original-edge coverage rigidity to obtain an injective map from every exact source fibre into $B_{\min}(G)$. Thus

$$|U_{S,p,r}| \leq 4094$$

for every exact type, and the actual cardinality-minimal blocker satisfies

$$|B^{(0)}| \leq 2048 \cdot 4094 = 8,384,512.$$

No smaller coverage-preserving blocker is introduced.

4.5. Reflected classical range

Lemma 4.9 (Reflected classical range). *Let $B = 8,384,512$. For every $n \leq B - 1$, the classical three-sunflower estimate implies*

$$|\mathcal{F}| \leq 2^n n! \leq (n+1)^n \leq B^n.$$

Proof. The standard Erdős–Rado recurrence gives $2^n n!$ for a three-sunflower-free n -uniform family [1]. By the arithmetic–geometric mean inequality,

$$n! \leq \left(\frac{n+1}{2}\right)^n.$$

Multiplication by 2^n gives $2^n n! \leq (n+1)^n$, and $n+1 \leq B$ in the stated range. $\square$

The terminal standing quotient is required only in the certificate-terminal high-rank branch $n \geq B$. In particular, the rank-two triangle regression is discharged before the terminal quotient is used in the main proof.

5. Collision disposition, terminal AASC closeout, and endpoint equivalence

5.1. Locked division of labour

Kernel Principle 5.1 (Governance classifies generated endpoints; it does not rewrite raw combinatorics). *The generated residual relation is taken literally. AASC does not replace a many-to-one residual map by an injective one, and kernel governance does not erase a collision that exists in*

the finite data. Governance is already forced once the generated carriers are determinate and non-degenerate. Its eliminative consequences are invoked after positive generation to determine which generated distinctions remain independently standing-bearing at the saturated endpoint once every lawful combinatorial disposition has been evaluated.

The distinction is essential. Generic anchor–tensor–skin classification says that a consequence-bearing difference is tensor rather than skin. It does not, by itself, say that the tensor difference is impossible. In the present proof the stronger conclusion comes from combining the finite collision-disposition theorem with certificate-terminal AASC exhaustion. A tensor difference must split, receive bounded charge, realize a sunflower, reconstruct through lower rank, change scope, or become impossible. At a fixed-scope terminal node all nonterminal alternatives have already been discharged.

5.2. The triangle regression

Proposition 5.2 (Rank-two triangle merge). *Let*

$$\mathcal{F}_\Delta = \{\{1, 2\}, \{1, 3\}, \{2, 3\}\}.$$

Then $\mathcal{F}_\Delta$ is three-sunflower-free, a cardinality-minimal blocker has two points, and the private deletions at its two roots produce the same singleton residual. Hence two distinct source roots reach one terminal coordinate.

Proof. No three distinct edges of the triangle have one common pairwise intersection, so the family contains no three-sunflower. Any two vertices hit all three edges, while one vertex misses the opposite edge; hence a minimum blocker has size two. Taking $B = \{1, 2\}$, private edges may be chosen as $\{1, 3\}$ for 1 and $\{2, 3\}$ for 2. Deleting the blocker root from each private edge gives the common residual $\{3\}$. $\square$

Corollary 5.3 (Unrestricted raw injectivity is false). *There is no theorem assigning every generated source root injectively to its raw terminal residual coordinate under only private-witness non-degeneracy, determinacy, and kernel applicability.*

Proposition 5.4 (The triangle has a lawful disposition). *The triangle collision is not an unresolved counterexample to terminal closeout. It enters the bounded residual-charge branch and is also contained in the reflected classical range of Lemma 4.9.*

Proof. The two residual parents form the permitted size-two residual fibre, so the collision is charged by the bounded same-side alternative rather than treated as a surviving independent terminal occupancy. Since its rank is two, the reflected classical estimate closes it before the high-rank certificate-terminal package is invoked. $\square$

Remark 5.5 (Exact regression meaning). The triangle permanently forbids the implication

$$\text{generated reachability} \implies \text{raw source-to-seed injectivity}.$$

It does not forbid

$$\begin{aligned} &\text{exhaustive disposition plus terminal closeout} \\ &\implies \text{injectivity on surviving coverage-bearing sources.} \end{aligned}$$

The latter is proved in Section 6.

5.3. Exhaustive collision disposition

Definition 5.6 (Generated collision). *A generated collision in an exact source fibre U is a pair of raw records*

$$a = (x, \pi, z), \quad b = (y, \sigma, z)$$

with one terminal locus z . It is endpoint-neutral when all admitted terminal probes agree. Otherwise its first differing admitted probe is consequence-bearing tensor content.

Theorem 5.7 (Generated collision disposition). *Every generated collision is assigned to exactly one of the following operational outcomes, with harmless overlaps resolved by the fixed priority order of the generated certificate:*

1. *endpoint-neutral skin or bookkeeping collapse;*
2. *bounded same-side residual charge;*
3. *tensor/profile separation or a change of comparison scope;*
4. *literal three-sunflower realization; or*
5. *strict lower-rank or endpoint-faithful factor reconstruction.*

No raw Hall premise is used in this classification.

Proof. The raw record contains its source, private-witness provenance, residual parent, next-incidence data, exact type, terminal locus, continuation profile, and literal certificate status. Compare two records in the fixed generated priority order. If no admitted coordinate differs, the multiplicity is endpoint-neutral. The permitted size-two residual fibre receives its bounded charge. A different finite profile, continuation target, or authorized domain separates the records or changes scope. A concrete triple with one common pairwise core yields the sunflower certificate. Every remaining endpoint-relevant collision carries an explicit reconstruction map and a strict decrease in the reduction potential. Finiteness makes the case split exhaustive. The record data generate the certificates; governance does not. □

Proposition 5.8 (Disposition discharge). *In the no-sunflower counter case:*

1. *endpoint-neutral collapse contributes one endpoint presentation;*
2. *bounded charge is paid by the already fixed finite residual allowance;*
3. *separated or scope-changed records do not remain dual occupants of one fixed comparison cell;*
4. *sunflower realization contradicts the counter case; and*
5. *strict reconstruction is discharged by well-founded induction on the reduction potential.*

5.4. Certified terminalization

Definition 5.9 (Finite counter case package). *A counter case package P records its rank n_P , family $\mathcal{F}_P$, active support X_P , generated structural ledger $\mathfrak{G}_P$, exact source fibres, raw terminal records, and every generated disposition certificate. It is determinate when these finite objects and their incidence maps are fixed, and non-degenerate when the private-witness data distinguish the generated source incidences.*

Definition 5.10 (Reduction potential). *Set*

$$\Phi(P) = (n_P, |\mathcal{F}_P|, |X_P|, |\mathfrak{G}_P|) \in \mathbb{N}^4$$

and order packages lexicographically.

Lemma 5.11 (Well-founded reduction). *The strict lexicographic order induced by Φ is well founded. Every certified lower-rank, proper-subfamily, support-restricted, or structurally simplified reconstruction decreases Φ .*

Proof. Lexicographic order on a finite product of well-founded orders on $\mathbb{N}$ is well founded. Each certificate records the first coordinate at which the reconstructed package is strictly smaller while all preceding coordinates agree. $\square$

Definition 5.12 (Endpoint-faithful split certificate). *A split certificate consists of finitely many packages Q_i with $\Phi(Q_i) < \Phi(P)$ together with reconstruction maps whose images cover the endpoint-relevant source data of P and preserve the recurrence-bearing incidence and coverage tensors.*

Definition 5.13 (Certificate-terminal package). *A package is certificate-terminal when every generated collision has been classified and none of the following remains unresolved in the fixed scope: literal sunflower realization, bounded same-side charge, tensor/profile separation, endpoint-faithful factorization, scope change, or strict lower-rank reconstruction. Any remaining multiplicity is endpoint-neutral presentation multiplicity unless the terminal AASC theorem declares the package impossible.*

Lemma 5.14 (Certified terminalization). *Every finite counterexample package admits a finite certified derivation tree whose internal nodes are disposition or reconstruction steps and whose leaves are certificate-terminal packages.*

Proof. At a nonterminal package, Theorem 5.7 either discharges the active collision directly or produces a certified split into packages with strictly smaller potential. Well-founded induction on Φ terminates every branch. Finite branching and finite source data yield a finite derivation tree. $\square$

Proposition 5.15 (Terminal exhaustion). *At a certificate-terminal leaf, no consequence-bearing tensor difference remains pending under the fixed comparison scope. If one remained, it would enter a nonterminal branch of Theorem 5.7. That contradicts terminality.*

Proposition 5.16 (Reduction-minimal counterexamples are terminal). *A counterexample minimal under Φ is certificate-terminal.*

Proof. Any nonterminal disposition would either close the package or produce a strictly smaller counterexample package, contradicting minimality. $\square$

5.5. Original-edge coverage rigidity

Definition 5.17 (Root coverage and terminal coverage tensor). *For $x \in B^{(0)}$, define*

$$\text{Cov}_{\mathcal{F}}(x) = \{E \in \mathcal{F} : x \in E\}.$$

For a raw terminal record $a = (x, \pi, z)$, define

$$\text{TCov}_{G,U}(a) = \text{Cov}_{\mathcal{F}}(x).$$

The tensor remembers which original edges are hit by the source root. It is not replaced by the terminal seed coordinate.

Proposition 5.18 (Coverage is endpoint tensor when load-bearing). *If changing the source coverage changes whether the realized source set hits an original edge, changes a link family, or changes reconstruction, then coverage is consequence-bearing tensor content and belongs to both endpoint signatures.*

Proof. Such a variation changes an admitted combinatorial consequence. By the anchor–tensor–skin classification, it cannot be skin. The terminal probe family and combinatorial signature therefore both retain it. $\square$

Theorem 5.19 (Minimal-blocker coverage antichain). *For $x, y \in B^{(0)}$,*

$$\text{Cov}_{\mathcal{F}}(x) \subseteq \text{Cov}_{\mathcal{F}}(y) \iff x = y.$$

In particular,

$$\text{Cov}_{\mathcal{F}}(x) = \text{Cov}_{\mathcal{F}}(y) \implies x = y.$$

Proof. The reverse implication is immediate. For the forward implication, let E_x be a private edge for x , so $E_x \cap B^{(0)} = \{x\}$. Since $E_x \in \text{Cov}_{\mathcal{F}}(x)$ and the coverage of x is contained in that of y , one has $y \in E_x$. Because $y \in B^{(0)}$, private-edge minimality forces $y = x$. $\square$

Corollary 5.20 (No strict coverage-preserving blocker representative). *If $D \subseteq B^{(0)}$ hits every edge of $\mathcal{F}$, then $D = B^{(0)}$. Equivalently, every strict subset of the cardinality-minimal blocker misses an original edge.*

Proof. If $x \in B^{(0)} \setminus D$, its private edge E_x meets $B^{(0)}$ only at x , so $D \cap E_x = \emptyset$. This contradicts the assumption that D is a blocker. $\square$

Corollary 5.21 (Coverage-respecting source equivalence is equality). *Any equivalence relation on $B^{(0)}$ whose related roots have equal original-edge coverage is the equality relation.*

Remark 5.22 (Correct quotient scope). Quotienting may remove several path histories over the same root, repeated ledger presentations, or other endpoint-neutral multiplicity. It cannot merge distinct coverage-bearing roots while retaining the blocker needed by the recurrence. A coverage-preserving representative blocker is therefore the surviving minimal blocker, not a smaller object.

5.6. Kernel necessity and terminal AASC exhaustion

The generated package is already finite, determinate, and private-witness non-degenerate. Its endpoint is used under a fixed comparison domain and an irreversible terminal report. These facts instantiate the kernel roles of Admissibility, Standing, Reference, and Irreversibility; they are not optional predicates added to a special subclass of sunflower families [15, 11, 18].

Kernel Principle 5.23 (Joint endpoint instantiation). *Standing, Reference, tensor content, coverage, and licensed metric description are jointly instantiated at the saturated endpoint. The logical restriction to a certificate-terminal package does not describe a temporal creation sequence. Metric readout remains non-authorizing: it counts the endpoint population but does not decide which differences have primitive standing [14, 9].*

Theorem 5.24 (Terminal AASC exhaustion). *Let P be a certificate-terminal, fixed-scope, non-degenerate determinate sunflower package in the no-three-sunflower branch. Suppose every generated bounded-charge, separation, scope-change, sunflower, factorization, and strict lower-rank*

alternative has been discharged. If two surviving raw terminal records occupy one terminal locus, then they have the same terminal standing class:

$$\text{seed}(a) = \text{seed}(b) \implies a \equiv_{\text{term}} b.$$

Equivalently, two independent standing-bearing endpoint occupants cannot remain at one fixed terminal locus.

Proof. The generated incidences have fixed carriers, faithful typed identity, private-witness non-degeneracy, a fixed comparison domain, and terminal use. Kernel necessity therefore supplies Admissibility, Standing, Reference, and Irreversibility. Consider two purported surviving presentations at one locus. A difference that changes no admitted consequence is skin and cannot support a second standing-bearing occupant. A consequence-bearing difference is tensor content by the anchor–tensor–skin classification [5, 8].

By Theorem 5.7, every such tensor difference must enter bounded charge, tensor/profile separation, scope change, sunflower realization, endpoint-faithful reconstruction, or strict lower-rank reconstruction. Proposition 5.15 says that none of those alternatives remains unresolved in the certificate-terminal fixed-scope package. Fixed-domain exhaustion excludes a second independent classifier on the same domain, Standing conservation forbids loss and retrospective recreation of an admitted endpoint distinction, and boundary-trace fixation prevents a discharged continuation distinction from re-entering at the final locus [13, 3, 12, 7, 16, 17].

Thus no consequence-bearing distinction remains that can sustain two independent occupants. The records agree under every terminal probe, which is precisely $a \equiv_{\text{term}} b$. $\square$

Corollary 5.25 (Terminal impossibility form). *Under the hypotheses of Theorem 5.24,*

$$a \not\equiv_{\text{term}} b \implies \text{seed}(a) \neq \text{seed}(b).$$

Remark 5.26 (Why generic ATS alone is insufficient). Generic ATS identifies a consequence-bearing difference as tensor. Theorem 5.24 is stronger because the finite proof has already exhausted every lawful continuation of that tensor difference. The impossibility conclusion belongs to the combined disposition-and-closeout architecture.

5.7. Two independently defined endpoint presentations

Theorem 5.27 (Terminal standing equivalence is final). *On a certificate-terminal package, $\equiv_{\text{term}}$ is an equivalence relation and every surviving terminal standing predicate, tensor, coverage coordinate, and endpoint status descends to $\text{AASCEnd}_{G,U}$.*

Proof. Equality under a fixed family of probes is reflexive, symmetric, and transitive. Each probe is constant on equivalence classes by definition. Theorem 5.24 supplies the fixed-locus occupancy rigidity needed for the terminal seed readout. $\square$

Proposition 5.28 (Combinatorial endpoint equivalence). *The relation $\equiv_{\text{comb}}$ is an equivalence relation, and every coordinate of $\text{csig}_{G,U}$ descends to $\text{CombEnd}_{G,U}$.*

Proof. It is equality of a finite generated signature. $\square$

Theorem 5.29 (Coverage descent). *If $a \equiv_{\text{term}} b$ or $a \equiv_{\text{comb}} b$, then*

$$\text{TCov}_{G,U}(a) = \text{TCov}_{G,U}(b).$$

Consequently there are well-defined coverage maps on both endpoint spaces.

Proof. Coverage is an admitted terminal probe and an explicit coordinate of the combinatorial signature. Equality in either relation therefore preserves it. $\square$

Theorem 5.30 (AASC–combinatorics identity biconditional). *For surviving terminal records in a certificate-terminal package,*

$$a \equiv_{\text{term}} b \iff a \equiv_{\text{comb}} b.$$

Proof. Suppose $a \equiv_{\text{term}} b$. Every coordinate of csig is an admitted terminal tensor, coverage, continuation, or literal status probe, so the signatures agree. Conversely, suppose $a \equiv_{\text{comb}} b$. The combinatorial signature fixes the terminal seed, private-edge incidence, original-edge coverage, continuation, reconstruction, factorization, exact type, and literal certificate status. Any remaining difference changes no admitted endpoint consequence and is therefore skin. Theorem 5.24 excludes an additional independent same-locus occupant, so all terminal probes agree and $a \equiv_{\text{term}} b$. $\square$

Theorem 5.31 (AASC–combinatorics endpoint equivalence). *The identity biconditional induces a unique endpoint-faithful equivalence*

$$\mathcal{B}_{G,U} : \text{AASCEnd}_{G,U} \xrightarrow{\sim} \text{CombEnd}_{G,U}$$

with inverse $\mathcal{B}_{G,U}^{-1}$, both represented by the identity on raw terminal records.

Proof. Two quotients of one set by extensionally equal equivalence relations are canonically equivalent. Theorem 5.30 gives the required equality of relations. $\square$

Corollary 5.32 (Bridge preservation laws). *The bridge and its inverse preserve terminal seed, private-edge incidence, original-edge coverage, continuation, reconstruction, factorization, exact type, and literal terminal status.*

Proof. Each readout descends from a coordinate shared by the two equivalent quotient presentations. $\square$

Remark 5.33 (Role of the bridge). The bridge is transfer after closeout. It expresses the same saturated endpoint under AASC and combinatorial presentations. It is not asked to prove terminal impossibility, and it does not create an injection that was absent from the terminal closeout theorem.

5.8. Quotient-final seed rigidity

Proposition 5.34 (Terminal seed maps are well defined). *The seed readout $\bar{\epsilon}_{G,U}$ is well defined on both $\text{AASCEnd}_{G,U}$ and $\text{CombEnd}_{G,U}$ and is preserved by $\mathcal{B}_{G,U}$.*

Proof. Seed equality is part of both endpoint relations. Bridge preservation follows from Corollary 5.32. $\square$

Theorem 5.35 (Quotient-final seed rigidity). *On the surviving certificate-terminal package, the terminal seed readout is injective:*

$$\bar{\epsilon}_{G,U}(C) = \bar{\epsilon}_{G,U}(D) \implies C = D.$$

This holds in either endpoint presentation.

Proof. Choose raw representatives a of C and b of D . Equal seed readouts place them at one terminal locus. Theorem 5.24 gives $a \equiv_{\text{term}} b$, hence $C = D$ in the AASC quotient. The endpoint equivalence transports the same conclusion to the combinatorial presentation. $\square$

Corollary 5.36 (Controlled endpoint population). *For every exact source fibre U ,*

$$|\text{AASCEnd}_{G,U}| = |\text{CombEnd}_{G,U}| \leq |B_{\min}(G)| \leq 4094.$$

Proof. Use Theorem 5.35 and the seed population bound of Theorem 4.1. $\square$

Audit Gate 5.37 (No blocker compression by endpoint quotienting). *Corollary 5.36 bounds endpoint classes. By itself it does not bound the number of source roots, and it does not construct a smaller blocker. Section 6 obtains the stronger source bound by combining seed rigidity with coverage descent and Theorem 5.19.*

6. Post-exhaustion source capacity and the actual minimal blocker

6.1. Selecting surviving endpoint records

Fix an exact source fibre $U = U_{S,p,r}$. By branch-complete path population, every $x \in U$ has at least one raw terminal record. Certified terminalization evaluates every generated branch, so in the surviving certificate-terminal package each source has at least one standing-bearing record to which the terminal closeout theorem applies.

Definition 6.1 (Terminal selector for a source fibre). *A terminal selector is any function*

$$\sigma_U : U \longrightarrow \text{RawTerm}_{G,U}$$

such that $\text{root}(\sigma_U(x)) = x$ for every $x \in U$. Define the selected endpoint and selected seed by

$$c_U(x) = [\sigma_U(x)] \in \text{AASCEnd}_{G,U}, \quad s_U(x) = \bar{\epsilon}_{G,U}(c_U(x)) \in B_{\min}(G).$$

Finite choice supplies such a selector. No canonical raw successor and no raw matching are assumed.

Proposition 6.2 (Selection-independent cardinal conclusion). *Every terminal selector produces an injective selected seed map s_U . Hence the cardinal estimate below is independent of which generated terminal record is chosen for each source.*

Proof. The injectivity proof in Theorem 6.4 uses only the source coverage carried by the selected record, quotient-final seed rigidity, and minimal-blocker coverage rigidity. Those properties hold for every admissible selection. $\square$

6.2. Endpoint equality returns to source equality

Lemma 6.3 (Selected endpoint equality preserves source coverage). *If $c_U(x) = c_U(y)$, then*

$$\text{Cov}_{\mathcal{F}}(x) = \text{Cov}_{\mathcal{F}}(y).$$

Proof. Equality of selected endpoint classes gives equality of their descended coverage tensors by Theorem 5.29. By Definition 5.17, the terminal coverage tensor of a selected record is exactly the original-edge coverage of its source root. $\square$

Theorem 6.4 (Post-exhaustion source-to-seed injectivity). *For every exact source fibre U , the selected seed map*

$$s_U : U \longrightarrow B_{\min}(G)$$

is injective.

Proof. Suppose $s_U(x) = s_U(y)$. By quotient-final seed rigidity, Theorem 5.35, the selected endpoint classes agree:

$$c_U(x) = c_U(y).$$

Lemma 6.3 then gives

$$\text{Cov}_{\mathcal{F}}(x) = \text{Cov}_{\mathcal{F}}(y).$$

Minimal-blocker coverage rigidity, Theorem 5.19, forces $x = y$. $\square$

Remark 6.5 (Exact relation to Hall). Theorem 6.4 has the finite-cardinality shape of a matching conclusion, but it is not raw Hall. The raw residual relation remains many-to-one. The injection is available only after the collision-disposition tree has been exhausted, terminal AASC closeout has made seed occupancy rigid, the endpoint bridge has transported that rigidity, and coverage has returned endpoint equality to equality of minimal-blocker roots.

Corollary 6.6 (Exact source-fibre capacity). *Every exact source fibre satisfies*

$$|U| \leq |B_{\min}(G)| \leq 4094.$$

Proof. The first inequality follows from Theorem 6.4; the second is Theorem 4.1. $\square$

Corollary 6.7 (No raw-fibre bound is assumed). *The estimate $|U| \leq 4094$ is a theorem of the terminal architecture, not a premise of positive generation. Before disposition, the proof assumes neither raw source injectivity nor raw history injectivity.*

6.3. The exact type partition

Let

$$\mathcal{T} = \text{im}(\text{supp}) \times \{0, 1\}^6 \times \{\text{Skin}, \text{BoundedFiber}, \text{TensorSplit}, \text{SunRole}\}.$$

By Theorem 4.5, $|\mathcal{T}| \leq 2048$. Since support, profile, and role are total functions, the associated source fibres form a disjoint partition:

$$B^{(0)} = \bigsqcup_{(S,p,r) \in \mathcal{T}} U_{S,p,r},$$

with empty fibres harmlessly included.

Theorem 6.8 (Actual minimal-blocker capacity). *The cardinality-minimal blocker satisfies*

$$|B^{(0)}| \leq 2048 \cdot 4094 = 8,384,512.$$

Proof. Using the exact type partition and Corollary 6.6,

$$\begin{aligned} |B^{(0)}| &= \sum_{(S,p,r) \in \mathcal{T}} |U_{S,p,r}| \\ &\leq |\mathcal{T}| \cdot 4094 \\ &\leq 2048 \cdot 4094 \\ &= 8,384,512. \end{aligned}$$

$\square$

Corollary 6.9 (The coverage-preserving representative is the minimal blocker). *Any subset of $B^{(0)}$ selected to preserve complete original-edge coverage is equal to $B^{(0)}$. Thus the blocker entering the recurrence is the actual bounded minimal blocker of Theorem 6.8.*

Proof. Apply Corollary 5.20. $\square$

6.4. Where quotienting remains useful

The quotient endpoint construction remains essential, but its scope is precise:

1. it identifies path histories that differ only by endpoint-neutral presentation multiplicity;
2. it gives a faithful AASC/combinatorial endpoint correspondence;
3. it supports a well-defined terminal seed readout and its post-closeout rigidity; and
4. it carries the original-edge coverage tensor unchanged.

It does not create a smaller coverage-bearing blocker. Once source injectivity has been derived, the cardinality readout applies to the surviving roots themselves.

Theorem 6.10 (Classical recurrence adequacy of the integrated bridge). *The combined closeout, bridge, and coverage-rigidity architecture supplies an actual blocker $B^{(0)}$ with*

$$|B^{(0)}| \leq 8,384,512.$$

No separate quotient-to-cardinality postulate is required.

Proof. Terminal closeout gives seed rigidity, the endpoint equivalence transfers it, Theorem 5.19 converts selected endpoint equality into source equality, and Theorem 6.8 bounds the actual minimal blocker. $\square$

7. Assembly of the endpoint theorem

7.1. One-point-link recurrence on the actual blocker

Let $f(n)$ denote the maximum size of a finite three-sunflower-free n -uniform family and put

$$B = 8,384,512.$$

For an actual blocker $D \subseteq X$ and $x \in D$, define the one-point link

$$\mathcal{F}(x) = \{E \setminus \{x\} : E \in \mathcal{F}, x \in E\}.$$

Each $\mathcal{F}(x)$ is $(n-1)$ -uniform and three-sunflower-free. Since every edge of $\mathcal{F}$ meets D ,

$$|\mathcal{F}| \leq \sum_{x \in D} |\mathcal{F}(x)| \leq |D|f(n-1).$$

Theorem 6.8 applies with $D = B^{(0)}$ and gives

$$f(n) \leq Bf(n-1).$$

The recurrence is therefore applied to the original cardinality-minimal blocker. No quotient representative substitutes for it.

7.2. Integrated proof

Proof of Theorem 1.9. Proceed by strong induction on n . The empty-rank case satisfies $f(0) \leq 1$. The reflected range $n \leq B - 1$ is Lemma 4.9.

Assume $n \geq B$ and suppose, for contradiction, that a counterexample exists. Choose a counterexample package minimal under the potential

$$\Phi(P) = (n_P, |\mathcal{F}_P|, |X_P|, |\mathfrak{G}_P|).$$

Track I supplies the cardinality-minimal blocker, private edges, residual tower, branch-complete terminal records, exact type partition, and every generated collision certificate. Theorem 5.7 assigns each collision to endpoint-neutral collapse, bounded charge, separation or scope change, sunflower realization, or strict reconstruction. Bounded and separated branches are discharged by their certificates, the sunflower branch contradicts the counterexample, and every reconstruction branch is settled by the induction hypothesis because it strictly decreases Φ .

By Proposition 5.16, the surviving fixed-scope package is certificate-terminal. Its generated incidences are determinate and private-witness non-degenerate, so kernel governance holds by necessity. The generated five-way classification and literal reduction certificates populate the terminal ledger; fixed-domain exhaustion removes the independent-authorizer branch, and Theorem 5.24 excludes two independent standing-bearing endpoint occupants at one terminal locus.

The terminal standing relation and the combinatorial endpoint relation then coincide by Theorem 5.30, and Theorem 5.31 gives

$$\text{AASCEnd}_{G,U} \simeq \text{CombEnd}_{G,U}$$

for every exact source fibre U . The bridge transports the closeout result, yielding quotient-final seed rigidity by Theorem 5.35.

Choose one surviving terminal record for each source root. If two selected roots receive the same terminal seed, seed rigidity makes their endpoint classes equal. Coverage descent then makes their original-edge coverage equal, and the minimal-blocker coverage antichain, Theorem 5.19, makes the roots equal. Hence the selected source-to-seed map is injective. Corollary 6.6 gives $|U| \leq 4094$ in every exact fibre, and the 2048-type partition gives

$$|B^{(0)}| \leq 2048 \cdot 4094 = B.$$

Applying the one-point-link recurrence to this actual blocker and using the induction hypothesis gives

$$|\mathcal{F}| \leq Bf(n-1) \leq B \cdot B^{n-1} = B^n,$$

contradicting the assumed counterexample. Therefore

$$|\mathcal{F}| \leq (8,384,512)^n.$$

□

7.3. Endpoint statement and proof status

The result is

$$\boxed{|\mathcal{F}| \leq (8,384,512)^n.}$$

The exponent applies to the complete parenthesized base. The proof preserves genuine raw merges, retains the AASC/combinatorial endpoint equivalence, and derives a post-exhaustion injection on the actual coverage-bearing source population. Its logical spine is

$$\begin{aligned} & \text{generation} \rightarrow \text{disposition} \rightarrow \text{certified terminalization} \\ & \text{AASC closeout} \rightarrow \text{endpoint equivalence} \rightarrow \text{seed rigidity} \\ & \text{coverage rigidity} \rightarrow \text{source injection} \rightarrow \text{minimal-blocker bound} \\ & \text{one-point-link recurrence.} \end{aligned}$$

The combined manuscript and Lean architecture therefore provides the complete hybrid AASC endpoint closure stated in the abstract. A separately formulated autonomous conventional replacement of the terminal AASC theorem would be a stronger single-language architecture, not an outstanding obligation within this theorem.

8. Lean correspondence and formal verification boundary

8.1. The controlling proof spine

The controlling formal account is release v0.4.0 of

AASC–Sunflower–Endpoint–Lean–Audit,

with the complete current project source snapshot included under `formal_spine/lean_sources/` [4]. The manuscript and the Lean development are complementary parts of one proof. Finite combinatorics remains the foreground construction. Kernel necessity supplies the structural authority on every generated determinate, non-degenerate endpoint. The manuscript-to-Lean boundary supplies only the Sunflower-specific generated five-way collision classification and literal reduction certificates.

Lean module	Verified role in the hybrid proof
GeneratedIncidenceTower	Positive residual and terminal-record generation; nonempty successor fibres; branch-complete incidence paths; preservation of the no-sunflower counterexample.
GeneratedTerminalKernel Governance	Faithful generated identity, private-witness non-degeneracy, endpoint preservation, and all four kernel roles on every generated terminal incidence.
MergeDispositionClosure	Real one-step and fine-fibre merge classification into sunflower realization, finite separation, bounded residual charge, or strict reconstruction.
ResidualMergeCounterexample	The rank-two triangle, its genuine many-to-one residual merge, failure of raw Hall, bounded-charge disposition, and automatic kernel governance without raw injection.

QuotientCoverageRigidity	Coverage-respecting equality, source-coordinate injectivity under an undischarged coverage bridge, coverage antichain, equality of any coverage-preserving representative blocker with the minimal blocker, and failure of strict representative subsets to hit.
ExplicitReductionPotential	Well-founded four-coordinate reduction, certified terminalization, evaluation of the derivation tree after terminal closeout, and endpoint consequences of closeout or impossibility.
TerminalRoleOccupancy Closeout	Bivalent skin/non-skin exhaustion, the explicit four-role identity ledger, certified non-skin branch discharge, terminal role injectivity, and fixed-identity realization.
KernelNecessityRole Occupancy	Necessity-rooted endpoint governance, exclusion of the attempted independent-authorizer branch, reduction-minimal terminality, four-role closeout, fixed identity, and the complete dependency trace.
ManuscriptKernelTransfer	The exact five-way witness-and-certificate import, kernel-necessity trace, canonical support-fibre bound, transfer to the sunflower theorem, and exact endpoint-strength correspondence.

8.2. Decisive declaration anchors

The controlling public names include:

```

GeneratedIncidenceTower.nextCoordinate_nonempty
GeneratedIncidenceTower.incidencePath_nonempty
GeneratedIncidenceTower.exists_reachesTerminalCarrier
GeneratedIncidenceTower.terminalFamily_noSunflower

MergeDispositionClosure.ThreePetalOneStepMergeDisposition
MergeDispositionClosure.FineFiberMergeDisposition

ResidualMergeCounterexample.triangle_oneStepPairMerge
ResidualMergeCounterexample.triangle_merge_has_boundedResidualCharge
ResidualMergeCounterexample.triangle_collision_is_nondegenerate_and_kernelGoverned

QuotientCoverageRigidity.CoverageRespectingSetoid.rel_iff_eq
QuotientCoverageRigidity.endpointCoverageSubsumes_iff_eq
QuotientCoverageRigidity.representativeBlocker_eq_minimalBlocker
QuotientCoverageRigidity.strictRepresentativeSubset_not_hitting

ExplicitReductionPotential.reductionPotential_wellFounded
ExplicitReductionPotential.certified_terminalization
ExplicitReductionPotential.CertifiedDerivation.settles_of_terminalCloseout
ExplicitReductionPotential.endpoint_of_exhaustion_and_terminalImpossibility

KernelNecessityRoleOccupancy.necessityRootedEndpointGovernance
KernelNecessityRoleOccupancy.NecessityRootedEndpointGovernance.noIndependentAuthorizer

```

```

KernelNecessityRoleOccupancy.KernelNecessityAnchoredFourRoleWitnessPipeline.
  fourRoleIdentityClosure
KernelNecessityRoleOccupancy.KernelNecessityAnchoredFourRoleWitnessPipeline.
  dependencyTrace

ManuscriptKernelTransfer.ThreePetalLocalManuscriptPopulation.kernelNecessityTrace
ManuscriptKernelTransfer.ThreePetalLocalManuscriptPopulation.canonicalSupportFiberBound
ManuscriptKernelTransfer.sunflower_of_importedManuscriptKernelGovernedClosure
ManuscriptKernelTransfer.ImportedManuscriptKernelGovernedPopulationTheorem.
  kernelNecessityEndpointTrace

```

The exact namespaces and source paths are recorded in `formal_spine/THEOREM_MAP.md`.

8.3. What is machine-checked

The local development verifies the following implication structure:

1. positive generated source and terminal population exists;
2. genuine residual collisions are allowed and classified rather than denied;
3. the triangle prevents any silent raw Hall assumption;
4. every strict reconstruction decreases the explicit reduction potential;
5. the finite certified derivation tree terminates;
6. every generated terminal incidence is determinate, non-degenerate, and governed by the kernel roles by necessity;
7. fixed-domain exhaustion excludes the fifth independent-authorizer candidate;
8. bivalent skin/non-skin exhaustion and certified branch discharge produce the four-role identity closeout;
9. fixed identity and the canonical support-fibre bound follow from that closeout;
10. coverage-preserving equivalence on minimal-blocker roots is equality; and
11. the manuscript witness-and-certificate import transfers through the endpoint theorem and exact endpoint correspondence.

In particular, the formal spine prevents both rejected inferences

$$\text{generated reachability} \implies \text{raw source injection}$$

and

$$\text{endpoint quotient} \implies \text{smaller coverage-preserving blocker}.$$

8.4. The exact manuscript-to-Lean boundary

The typed boundary is `ImportedManuscriptKernelGovernedPopulationTheorem`. Its local output, `ThreePetalLocalManuscriptPopulation`, contains:

1. the generated five-way candidate ledger: skin, bounded certificate, tensor split, sunflower, or attempted independent authorizer; and
2. literal reduction-minimal certificates for the non-skin branches.

It does not contain a `kernelApplies` field, a raw Hall selector, the support-fibre bound, fixed identity, or the endpoint theorem. Lean constructs necessity-rooted governance from the generated determinate endpoint, removes the fifth candidate by the fixed-domain kernel theorem, derives certificate terminality and the four-role identity closeout, and then derives fixed identity, the fibre bound, and the endpoint.

Audit Gate 8.1 (No hidden endpoint assumption). *The positive generation and disposition modules do not assume the final blocker bound, a raw matching, one occupant per token, or an injective source code. The manuscript boundary is exposed at the target-specific witness-and-certificate level, and every downstream structural and cardinal consequence is checked from its exact type.*

8.5. Verification record

The supplied verification packet records:

- Lean toolchain `leanprover/lean4:v4.28.0`;
- successful public-entrypoint build: 1,143 jobs;
- successful combined entrypoint-and-audit build: 1,146 jobs;
- 250 focused `#print axioms` checks;
- no live project declaration using `axiom`, `sorry`, `admit`, or an unsafe escape; and
- only ordinary Lean/Mathlib principles such as `Classical.choice`, `propext`, and `Quot.sound` on the wider audited surface.

The bundled source snapshot contains every project `.lean` file used by the release, while omitting only external dependencies and generated caches. `SOURCE_HASHES.txt` fixes every included source identity.

8.6. Autonomous conventional strengthening not required by the theorem

The release also contains exploratory modules relevant to a distinct single-language strengthening:

- `RankCompatibleProductGate`;
- `GradedProductConvolution`; and
- `RankWeightedMergeCharging`.

These modules correct unrestricted product fullness, isolate same-rank missing tuples, construct strict positive reconstructions, and bound fixed-rank slices. They are not load-bearing components of the theorem proved here. A separate question asks whether the terminal AASC impossibility theorem can be replaced entirely by autonomous conventional reconstruction. That question concerns a different proof architecture; its resolution is neither assumed nor required by the integrated closure.

8.7. Exact status statement

The manuscript and Lean together provide the complete integrated hybrid endpoint closure. It applies to every non-degenerate determinate sunflower package. Lean formalizes the combinatorial and proof-theoretic component at its natural boundary: positive

generation, merge disposition, the triangle regression, coverage rigidity, certified terminalization, the necessity-rooted kernel packet, elimination from five candidates to four roles, fixed identity, the support-fibre bound, and endpoint transfer. The manuscript proves the generated five-way collision classification and its literal reduction certificates at their natural boundary. The typed import joins those completed components. No autonomous conventional endpoint is required by the theorem’s statement or dependency structure.

9. Adversarial audit

9.1. The triangle disproves all source injectivity

Attack. Two triangle roots reach one terminal singleton, so no root-to-seed injection can occur anywhere in the proof.

Answer. The triangle disproves unrestricted pre-discharge injectivity. Its collision enters bounded residual charge and the reflected low-rank closeout. Theorem 6.4 applies only after exhaustive disposition and certificate-terminal AASC closeout. It does not alter the raw triangle map.

9.2. Coverage-preserving quotienting creates a smaller blocker

Attack. Choose one source representative from each endpoint class and use the resulting smaller set as the blocker.

Answer. This inference is false. Theorem 5.19 proves that original-edge coverage profiles of minimal-blocker roots form an antichain. Corollary 5.20 proves that every proper subset of $B^{(0)}$ misses a private edge. Quotienting may reduce duplicate histories; a coverage-preserving blocker representative equals $B^{(0)}$.

9.3. Generic ATS does not prove seed injectivity

Attack. Anchor–tensor–skin classification only says that an admitted difference is tensor. It does not imply one endpoint occupant per seed.

Answer. Correct as a statement about generic ATS alone. The proof uses the stronger certificate-terminal AASC exhaustion theorem. Before closeout, every tensor difference has already been assigned to bounded charge, separation, scope change, sunflower realization, factorization, or strict lower-rank reconstruction. At a fixed terminal node none of those alternatives remains. The impossibility of dual occupancy follows from that combined exhaustion, not from generic classification by itself.

9.4. The bridge is being asked to create injectivity

Attack. The equivalence $\text{AASCEnd} \simeq \text{CombEnd}$ is circular because seed injectivity is needed to establish it and is then read back from it.

Answer. The dependency is one-way and explicit. Terminal AASC closeout supplies same-locus occupancy rigidity first. The biconditional then identifies the AASC and combinatorial endpoint relations and transports the already established closeout. Quotient-final seed rigidity follows. Coverage rigidity is a separate minimal-blocker theorem and converts endpoint equality into source equality. No step cites its own conclusion.

9.5. The proof silently assumes Hall

Attack. The final injection is merely Hall under another name.

Answer. No Hall condition is assumed for the raw reachability relation. The triangle regression formally prevents that. The selected source map is constructed only after terminal closeout; any selection of one generated terminal record per source is injective because equal selected seeds imply equal endpoint classes and then equal coverage-bearing roots. The resulting finite injection is a downstream Hall-like corollary, not a premise or a matching theorem for raw residualization.

9.6. The source selector is not canonical

Attack. A source may have several generated terminal histories, so the cardinal map is ill defined.

Answer. The proof does not require a canonical raw successor. Finite choice selects one terminal record for each source. Proposition 6.2 proves that every such selection yields an injective seed map. The cardinal conclusion is therefore choice-independent even though the raw history selection need not be canonical.

9.7. Coverage is added after the endpoint is formed

Attack. The proof first quotients records and then pastes original-edge coverage back onto the classes.

Answer. Coverage is an admitted terminal probe and an explicit coordinate of csig from the outset. Theorem 5.29 states its descent through both endpoint presentations. It is not reconstructed from cardinality and cannot be discarded when it changes the blocker or recurrence.

9.8. Metricity is temporally later than Standing

Attack. The proof requires an ontological sequence in which Standing exists before any metric description can coexist with it.

Answer. No temporal succession is claimed. Standing, Reference, tensor content, coverage, and licensed metric description are jointly instantiated in the saturated endpoint. Metricity is non-authorizing: it reports a population whose identity constraints are already fixed, but does not generate those identities.

9.9. The formal import makes the theorem conditional

Attack. Because Lean imports part of the manuscript theorem rather than deriving it from PDF text, the mathematical endpoint is conditional.

Answer. The import marks a machine-internalization boundary, not an optional family-side hypothesis. The imported type now contains only the Sunflower-specific generated five-way collision classification and literal reduction certificates. Lean independently constructs necessity-rooted kernel governance, eliminates the attempted independent-authorizer branch, derives certificate terminality and the four-role identity closeout, obtains fixed identity and the support-fibre bound, and transfers those results to the endpoint. A named formal boundary does not make kernel governance elective or weaken the manuscript proof.

9.10. A stronger single-language theorem is required for this result

Attack. Unless the terminal AASC theorem is also replaced by an autonomous conventional reconstruction, the present endpoint remains incomplete.

Answer. This confuses two proof classes. The theorem asserted here integrates combinatorics with AASC. Its combinatorial component supplies positive generation, lawful merge disposition, well-founded reduction, certificate terminality, coverage rigidity, and endpoint transfer; its AASC component supplies the forced terminal impossibility theorem. Both components are complete in their assigned roles and are joined by a typed interface. Replacing the terminal theorem by autonomous conventional reconstruction would produce a distinct stronger architecture, not complete a missing step in this one.

9.11. The typed coordinates hide uncounted multiplicity

Attack. Support, profile, role, and terminal seed omit other endpoint distinctions, so the 2048·4094 count is incomplete.

Answer. The finite type coordinates partition sources; they are not claimed to exhaust all tensor content. Full continuation, coverage, reconstruction, and status information remains in the endpoint relation used to establish seed rigidity. Once source-to-seed injectivity is proved within each exact type, every source is charged exactly once by its type and seed. An additional standing-neutral label does not create another source; a standing-relevant difference must enter the disposition and endpoint signature before closeout.

9.12. The reduction can recurse indefinitely

Attack. Repeated split or reconstruction branches may never reach the terminal package.

Answer. Every certified recursive branch strictly decreases

$$\Phi(P) = (n_P, |\mathcal{F}_P|, |X_P|, |\mathfrak{G}_P|)$$

lexicographically. Lemma 5.11 and Lemma 5.14 give a finite derivation tree.

9.13. Hostile acceptance test

A valid implementation must answer all of the following affirmatively:

1. Are private witnesses, residuals, paths, and terminal records positively generated before AASC use?
2. Does the triangle remain a counterexample to unrestricted raw Hall?
3. Is every collision assigned to an explicit lawful disposition?
4. Does every recursive disposition strictly decrease the reduction potential?
5. Is the certificate-terminal AASC exhaustion theorem stated before seed rigidity?
6. Are $\equiv_{\text{term}}$ and $\equiv_{\text{comb}}$ defined independently and proved equivalent after closeout?
7. Is original-edge coverage retained in both endpoint presentations?
8. Does coverage rigidity prevent distinct minimal-blocker roots from collapsing?
9. Is $|U| \leq 4094$ derived rather than assumed?

10. Is the bounded object the actual minimal blocker $B^{(0)}$?
11. Is the one-point-link recurrence applied directly to $B^{(0)}$?
12. Is the Lean witness-and-certificate import boundary distinguished from the mathematical theorem?
13. Is an autonomous conventional replacement correctly identified as a distinct strengthening rather than an unfinished part of the hybrid proof?

The integrated edition satisfies all thirteen gates.

10. Appendices and audit ledgers

10.1. Generated, endpoint, and source populations

Object	Status and role
Source root $x \in U$	An actual coordinate of the cardinality-minimal blocker. Distinct roots have incomparable original-edge coverage.
Raw terminal record (x, π, z)	A positively generated history with source, path, terminal seed, coverage, continuation, and certificate data.
Raw merge fibre	All records reaching one terminal seed. It may contain records from distinct roots before disposition.
Endpoint-neutral history quotient	Removes route order, duplicate ledger entries, and other presentation multiplicity that changes no admitted endpoint consequence.
AASC relation $\equiv_{\text{term}}$	Equality under all terminal Standing, Reference, tensor, coverage, and status probes.
AASC endpoint $\text{AASCEnd}_{G,U}$	The terminal standing presentation after certificate-terminal closeout.
Combinatorial relation $\equiv_{\text{comb}}$	Equality of the generated recurrence-sufficient terminal signature.
Combinatorial endpoint $\text{CombEnd}_{G,U}$	The faithful generated presentation carrying seed, incidence, coverage, continuation, reconstruction, and status data.
Bridge $\mathcal{B}_{G,U}$	The unique equivalence between the two endpoint presentations after closeout.
Selected source map s_U	Any post-closeout selection of one terminal record per source followed by terminal seed readout; every such map is injective.
Actual blocker $B^{(0)}$	The bounded minimal blocker used in the recurrence. Every coverage-preserving source subset is the complete minimal blocker.

10.2. Collision-disposition ledger

Disposition	Terminal effect
-------------	-----------------

Endpoint-neutral skin or bookkeeping collapse	Several histories represent one endpoint occupant.
Bounded same-side residual charge	The collision is paid by the fixed finite residual allowance and does not remain an uncharged dual occupancy.
Tensor/profile separation	The records lie in different exact comparison cells.
Scope change	The records are no longer rival occupants under one fixed domain.
Three-sunflower realization	Contradicts the no-sunflower counterexample.
Endpoint-faithful factor reconstruction	Reconstructs the endpoint from certified smaller factors.
Strict lower-rank reconstruction	Settled by induction under a strict decrease of Φ .
Terminal impossibility	After all other branches are discharged, AASC excludes independent same-locus occupancy.

10.3. Endpoint bridge commuting ledger

For $C \in \text{AASCEnd}_{G,U}$ and $K = \mathcal{B}_{G,U}(C)$, the readouts commute:

$$\begin{aligned}
\epsilon_{G,U}^{\text{comb}}(K) &= \bar{\epsilon}_{G,U}(C), \\
\text{Cov}_{G,U}^{\text{comb}}(K) &= \overline{\text{TCov}_{G,U}}(C), \\
\text{Inc}^{\text{comb}}(K) &= \text{Inc}^{\text{AASC}}(C), \\
\text{Cont}^{\text{comb}}(K) &= \text{Cont}^{\text{AASC}}(C), \\
\text{Reconstruct}^{\text{comb}}(K) &= \text{Reconstruct}^{\text{AASC}}(C), \\
\text{Status}^{\text{comb}}(K) &= \text{Status}^{\text{AASC}}(C).
\end{aligned}$$

The bridge is endpoint-faithful, not merely cardinality-preserving.

10.4. Coverage rigidity ledger

For $x, y \in B^{(0)}$:

$$\text{Cov}_{\mathcal{F}}(x) \subseteq \text{Cov}_{\mathcal{F}}(y) \iff x = y.$$

The implications used in the source-capacity proof are:

$$\begin{aligned}
&s_U(x) = s_U(y) \\
&\Downarrow \text{ quotient-final seed rigidity} \\
&c_U(x) = c_U(y) \\
&\Downarrow \text{ coverage descent} \\
&\text{Cov}_{\mathcal{F}}(x) = \text{Cov}_{\mathcal{F}}(y) \\
&\Downarrow \text{ private-edge minimality} \\
&x = y.
\end{aligned}$$

A source representative set preserving all original-edge coverage is therefore $B^{(0)}$ itself.

10.5. Simultaneity and authority ledger

Feature	Correct interpretation
Standing and Reference	Jointly instantiated kernel roles on the determinate endpoint package.
Tensor content and coverage	Jointly instantiated consequence-bearing coordinates; neither is pasted on after quotienting.
Metric description	A licensed concurrent readout of the saturated endpoint population.
Cardinality	Reports how many endpoint or source objects survive the constraints; it does not generate identity.
“Post-exhaustion”	A logical restriction to terminal hypotheses, not a temporal or ontological sequence.
Historical multiplicity	Countable as descriptions but non-authorizing when endpoint-neutral.

10.6. Triangle regression certificate

The rank-two triangle must remain in every proof and software regression suite. A modification fails the regression if it proves any of the following without the certificate-terminal hypotheses:

- raw terminal injectivity;
- raw reachable Hall;
- one raw root per seed; or
- disappearance of the genuine many-to-one residual merge.

The required outputs are:

- two distinct minimal-blocker roots;
- one common residual singleton;
- failure of the premature source bridge;
- bounded-charge disposition; and
- reflected low-rank closeout.

10.7. Reduction-potential comparison table

Reduction type	Required strict decrease
Rank deletion	$n_Q < n_P$
Same-rank proper subfamily	$n_Q = n_P$ and $ \mathcal{F}_Q < \mathcal{F}_P $
Same-rank, same-count support restriction	Preceding coordinates agree and $ X_Q < X_P $
Same numerical package with simplified generated structure	Preceding coordinates agree and $ \mathfrak{G}_Q < \mathfrak{G}_P $

No coordinate decreases

Not a certified reconstruction

10.8. Dependency ledger

Layer	Status
Minimal blocker and private witnesses	Proved by finite combinatorics
Residual tower and branch-complete paths	Proved by finite combinatorics
Raw residual merge existence	Retained and witnessed by the triangle
Collision disposition	Generated finite classification
Certified terminalization	Proved by well-founded induction on Φ
Kernel applicability	Necessary on determinate, non-degenerate endpoint incidence
Sunflower-specific witness population	Manuscript theorem supplies the generated five-way classification and literal reduction certificates
Terminal AASC exhaustion/impossibility	Derived in Lean from kernel necessity, fixed-domain fifth-branch exclusion, four-role exhaustion, and certificate terminality
AASC terminal relation	Defined independently by admitted endpoint probes
Combinatorial endpoint relation	Defined independently by the generated endpoint signature
Identity biconditional	Proves the two relations coincide after closeout
Endpoint equivalence	Transfers the saturated endpoint between faithful presentations
Quotient-final seed rigidity	Consequence of terminal closeout and bridge transfer
Minimal-blocker coverage rigidity	Proved by private-edge minimality
Source-to-seed injectivity	Derived after closeout from seed and coverage rigidity
Exact source-fibre capacity	$ U \leq 4094$
Actual minimal-blocker capacity	$ B^{(0)} \leq 8,384,512$
Endpoint	$ \mathcal{F} \leq (8,384,512)^n$

10.9. Formalization ledger

The local Lean development checks:

- positive residual and terminal-record generation;
- the triangle merge regression and bounded-charge disposition;
- the bounded/separated/sunflower/lower-rank collision architecture;
- minimal-blocker coverage rigidity and representative-blocker equality;
- well-founded certified terminalization;
- necessity-rooted kernel governance on every generated terminal incidence;
- elimination of the attempted independent-authorizer branch;
- four-role identity closeout, fixed identity, and the support-fibre bound;

- the generic endpoint implication from terminal closeout or impossibility; and
- the manuscript witness-and-certificate transfer and exact endpoint correspondence.

The manuscript boundary supplies only the Sunflower-specific five-way generated classification and literal reduction certificates. Lean derives terminality, the four-role closeout, fixed identity, the fibre bound, and the endpoint from that boundary. The complete current Lean source snapshot and verification record are bundled under `formal_spine/`.

10.10. Prohibited interpretation drift

The following substitutions are incompatible with the theorem package:

- replacing the generated many-to-one residual relation by a raw injective matching;
- treating generic ATS classification as the terminal exhaustion theorem;
- asking the endpoint equivalence to prove closeout by itself;
- quotienting away original-edge coverage or continuation tensors;
- claiming that a coverage-preserving quotient creates a smaller blocker;
- treating post-exhaustion source injectivity as an assumed raw Hall condition;
- describing metricity as temporally absent while Standing is present;
- treating the manuscript and Lean development as competing solutions; or
- treating an autonomous conventional replacement as an unfinished condition on the integrated theorem.

10.11. Extraction-robust formulas

The recurrence is displayed as

$$f(n) \leq B f(n-1) \leq B \cdot B^{n-1} = B^n.$$

The endpoint is written

$$(8,384,512)^n,$$

so the exponent remains attached to the complete parenthesized base under imperfect text extraction. ASCII fallbacks are:

```
f(n) <= B f(n-1) <= B * B^(n-1) = B^n
|U| <= 4094
|B^(0)| <= 2048 * 4094 = 8,384,512
|F| <= (8,384,512)^n
```

10.12. Publication-status statement

This integrated edition presents the hybrid proof in its controlling order: positive generation, target-specific five-way witness population, certified terminalization, necessity-rooted four-role AASC closeout, endpoint equivalence, quotient-final seed rigidity, minimal-blocker coverage rigidity, post-exhaustion source injection, actual blocker capacity, and the classical recurrence. Raw merges remain literal, quotienting is restricted to endpoint-neutral histories, and the recurrence is applied to the actual minimal blocker. The mathematical and formalization boundaries are stated explicitly and consistently throughout the manuscript and release reports.

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